

Master of Science (M.Sc.)

”Mannheim Master in Data Science“



– Module catalog –
for students starting in or after fall 2024

Academic Year
HWS 2026 / FSS 2027

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Foreword

This document describes the courses that will be offered in HWS 2026/ FSS 2027 for students studying M. Sc. Mannheim Master in Data Science (Examination Regulations for the Master's program from 20 th February 2024). You can find the Examination Regulations on the website of the Student Services (Studienbüros):

<https://www.uni-mannheim.de/en/academics/during-your-studies/examinations/examination-regulations/>

It is possible that additional courses will be made available during the course of the academic year. These will be published in an appendix available on the following web page:

<https://www.wim.uni-mannheim.de/studium/studienorganisation/mannheim-master-in-data-science-beginn-hws-2024-25/>

A. Overview

		ECTS
Data Science Fundamentals	Four “Data Science Fundamentals” with 27 ECTS	27
Data Management	Minimum of one “Data Management” course	6-24
Data Analytics Methods	Minimum of two “Data Analytics Methods” courses with at least 12 ECTS	12-36
Responsible Data Science	Minimum of one “Responsible Data Science” course	3-7
Data Science Applications	Up to 12 ECTS in elective courses	0-12
Projects and Seminars	Team Project or Individual Project, Scientific Research and Seminars	14-18
Master’s Thesis	Six-months-long written academic assignment	30
Total		120

Abbreviations:

- HWS (Herbst-/Wintersemester): Course is offered in the respective Fall semester.
- FSS (Frühjahrs-/Sommersemester): Course is offered in the respective Spring semester.
- HWS/FSS: Course is offered both in Fall semester and Spring semester.

B. Data Science Fundamentals

1. Overview

Module no.	Name of Module	Offered	Language	ECTS	Page
CS 560	Large Scale Data Management	HWS	E	6	5
CS 652	Data Security and Privacy	FSS*	E	6	7
IE 500	Data Mining	HWS/FSS	E	6	9
IE 675b	Machine Learning	HWS	E	9	11

2. Detailed descriptions

CS 560: Large-Scale Data Management

Module number	CS 560
Title	Large-Scale Data Management
Form of module	Lecture with exercise
Type of module	Computer Science Fundamental
Level	Master
ECTS	6
Workload	Hours in presence: 42 h per semester (2+2 SWS) Exercises and self-study: 119 h per semester
Prerequisites	Database systems, algorithms & data structures, Java/Python programming
Aim of module	This course introduces the fundamental concepts and computational paradigms of large-scale data management and Big Data. This includes methods for storing, updating, querying and analyzing large dataset as well as for data-intensive computing. The course covers concept, algorithms and system issues; accompanying exercises provide hands-on experience. Topics include: • Parallel and distributed databases • Parallel database design • Distributed query processing • Big data processing (including MapReduce, Spark, Streaming) • Distributed transactions • NoSQL databases
Learning outcomes and qualification goals	Professional competence: • Students will acquire knowledge about methods and systems for working with large datasets and data-intensive computing (MK1, MK2) Methodological competence: • Be able to judge, select and use traditional and modern data management systems for a given data intensive task • Be able to solve computational problems involving large datasets (MF1) Personal competence: • Study independently (MKO3)
Media	Slide set, black board, exercise sheets, datasets, software
Literature	• T. Özsu, P. Valduriez: <i>Principles of Distributed Database Systems</i>, Springer, 4th ed., 2020 • H. Garcia-Molina, J. D. Ullman, J. Widom: <i>Database Systems: The Complete Book</i>, Prentice Hall, 2nd ed., 2008 • More in lecture notes
Methods	Lecture, weekly exercise, experimentation with different systems

Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr. Rainer Gemulla
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	1st/ 2nd semester

CS 652: Data Security and Privacy

Module number	CS 652
Title	Data Security and Privacy
Form of module	Inverted classroom with exercises
Type of module	Fundamental in Computer Science
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 112 h per semester
Prerequisites	There are no formal prerequisites but knowledge in cryptography or IT-security is recommended, e.g., by attending the lectures “Kryptographie I” or “Selected Topics in IT-Security”
Aim of module	Nowadays, more and more user-related data is collected and processed – either willingly as in the context of Facebook or Payback or possibly unconsciously as in the case of the Internet of Things. The aim of the module is to raise awareness, in particular with respect to privacy risks. This is done by showing various threats. Possible examples are traces left on the Internet such as the use of cookies or browser fingerprinting but also leakage in the context of machine learning. In addition, a variety of privacy enhancing technologies will be discussed, possibly such as anonymization of data, differential privacy, or the use of dedicated cryptographic schemes
Learning outcomes and qualification goals	Professional competence: • Students will acquire the knowledge to identify security and privacy threats and to select and use appropriate countermeasures (MK2) Methodological competence: • Successful participants will be able to understand, to select, apply and evaluate the most appropriate techniques for a variety of different privacy-sensitive scenarios • In particular they are able to realize possible risks in new scenarios and to transfer given solutions to these (MK1) Personal competence: • The analytic, concentrated and precise thinking of the students is trained • By the independent treatment of applications, e.g. in the course of the exercises, their abstraction capacity is further developed and the transfer of the learned material to related questions is trained (MF1, MKO3)
Media	Video recordings, annotated lecture slides
Literature	Will be announced in the lecture

Methods	Reworking the lecture and studying the relevant literature in self-study. During the lecture: discussing questions and ideas and working together on concrete examples. Solving exercises in self-study and in practice in cooperation with fellow students
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	FSS
Person in charge	Prof. Dr. Frederik Armknecht
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science, M.Sc. Business Informatics, Teacher Education Informatics, M.Sc. Mathematics
Semester	1st/ 2nd/ 3rd semester

IE 500: Data Mining

Module number	IE 500
Title	Data Mining
Form of module	Lecture with exercise and project
Type of module	Business Informatics Fundamental
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 98 h per semester, including • 70 h: Pre- and post-lecture studying and revision • 28 h: Examination preparation
Prerequisites	Foundations of Statistics, Practical Informatics I
Aim of module	The course provides an introduction to advanced data analysis techniques as a basis for analyzing business data and providing input for decision support systems. The course will cover the following topics: • Goals and Principles of Data Mining • Data Representation and Preprocessing • Classification • Regression • Cluster Analysis • Association Analysis • Practical Data Mining using Python
Learning outcomes and qualification goals	Professional competence: • Students will acquire basic knowledge of the techniques, opportunities and applications of data mining (MK1, MF1) Methodological competence: • Successful participants will be able to identify opportunities for applying data mining in an enterprise environment, select and apply appropriate techniques, and interpret the results • Project organization skills (MK2, MF3, MF4, MKO1) Personal competence: • Team work skills • Presentation skills (MKO2, MF2)
Media	Slide set, exercise sheets, Jupyter notebooks, data sets for the exercises, lecture videos
Literature	• Pang-Ning Tan, Michael Steinback, Vipin Kumar: <i>Introduction to Data Mining</i>, 2nd Edition, Pearson • Aurélien Géron: <i>Hands-on Machine Learning with Scikit-Learn, Keras & TensorFlow</i>, 3rd Edition, O'Reilly

Methods	The course consists of a lecture together with accompanying practical exercises as well as student team projects. In the exercises the participants will gather initial expertise in applying state of the art data mining tools on realistic data sets. The group projects take place in the last third of the term. Within the projects, students implement advanced data mining projects of personal choice and report about the results of their projects in the form of a written report as well as an oral presentation
Form of assessment	• Written examination (75%) • Project report (20%) • Oral project presentation (5%)
Admission requirements for assessment	-
Duration of assessment	• 60 minutes (written examination) • 5-11 pages (project report) • 5-15 minutes (oral project presentation)
Language	English
Offering	HWS/FSS
Person in charge	Prof. Dr. Heiko Paulheim, Prof. Dr. Christian Bizer, Dr. Sven Hertling
Duration of module	1 semester
Further modules	• IE 675b - Machine Learning • IE 678 - Deep Learning • IS 661 (IS 6610) - Text Analytics • CS 646 - Higher Level Computer Vision
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	1st/ 2nd semester

IE 675b: Machine Learning

Module number	IE 675b
Title	Machine Learning
Form of module	Lecture with exercise
Type of module	Fundamental course (MMDS), Specialization Course (M.Sc. Wifo)
Level	Master
ECTS	9
Workload	Hours in presence: 63 h per semester (4+2 SWS) Exercises, assignments, and self-study: 167 h per semester
Prerequisites	IE 500 Data Mining I (recommended), probability/linear algebra
Aim of module	Machine learning is concerned with building computer systems that improve with experience as well as the study of learning processes, including the design of algorithms that are able to make predictions or extract knowledge from data. The aim of this module is to provide an introduction into the field of machine learning, and study algorithms, underlying concepts, and theoretical principles. Topics include: • Basics of machine learning • Estimators, inference (MLE, ERM, RRM as well as MAP/Bayesian inference) and decision • Selected classification and regression models • Parameter estimation (MLE, MAP, ERM, RRM, Bayesian, first- and second-order methods) • Dimensionality reduction (SVD, PCA, FA) • Expectation maximization and mixture models • Kernels, kernel trick, vector machines and sparsity • Hyperparameter optimization
Learning outcomes and qualification goals	Professional competence: • Deep understanding of algorithms and underlying concepts of machine learning (MK1, MF1) Methodological competence: • Being able to apply machine learning techniques and systems for a given problem • Being able to model and implement new machine learning techniques (MK2, MF3, MF4) Personal competence: • Writing skills • Presentation skills • Statistical programming skills (MKO3, MF2)
Media	Slide set, exercise sheets, software, datasets
Literature	• K.P. Murphy. <i>Probabilistic Machine Learning: An Introduction</i>. The MIT Press, 2022 • Additional material, books and articles provided in lecture notes

Methods	The course consists of a lecture accompanied by theoretical and practical exercises as well as case studies with real data. In the exercises, students will deepen the material discussed in the lecture, apply the methods in practice and present the result
Form of assessment	Written examination
Admission requirements for assessment	Homework assignments (pass at least 3 assignments)
Duration of assessment	120 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr. Rainer Gemulla
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	1st/ 2nd/ 3rd semester

C. Data Management

1. Overview

Module no.	Name of Module	Offered	Language	ECTS	Page
AC 651	Additional Course - Data Management	HWS/FSS	E	Max. 18	14
CS 500	Advanced Software Engineering	HWS	E	6	15
CS 550	Algorithmics	FSS	E	6	17
CS 600	Model-driven Development	HWS	E	6	19
CS 606	Foundations of Artificial Intelligence: Search and Problem Solving	HWS	E	6	21
CS 651	Cryptography II	HWS	E	6	23
CS 664	Blockchain Security	HWS	E	6	25
IE 650	Knowledge Graphs	HWS	E	6	27
IE 670	Web Data Integration	HWS	E	3	29
IE 683	Web Data Integration Project	HWS	E	3	31
DS 200	Sampling and Data	HWS	E	9	33

* For a detailed description, please see the module catalogue of the respective following degree programs:

- MMM: M.Sc. Mannheim Master in Management,
<https://www.bwl.uni-mannheim.de/en/module-catalog-mmm/>

2. Detailed descriptions

AC 651: Additional Course - Data Management

Module number	AC 651
Title	Additional Course - Data Management
Form of module	Depends on course
Type of module	Depends on course
Level	Master
ECTS	Max. 18 (A total of 18 ECTS can be recognized over all possible additional courses)
Workload	Depends on course
Prerequisites	Depends on course
Aim of module	The course falls into the data management area of the MMDS and covers topics related to data management but is not directly equivalent to any course in the MMDS module catalogue. The course level equals a regular course in MMDS study program. The module can be taken either at the University of Mannheim or at any other university in Germany or abroad
Learning outcomes and qualification goals	Depends on course
Media	Depends on course
Literature	Depends on course
Methods	Depends on course
Form of assessment	Depends on course
Admission requirements for assessment	Depends on course
Duration of assessment	Depends on course
Language	English preferred, but any other language possible if Mannheim faculty member is able to identify content and level
Offering	HWS/FSS
Person in charge	Lecturer at the host university
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science
Semester	2nd/ 3rd/ 4th semester

CS 500: Advanced Software Engineering

Module number	CS 500
Title	Advanced Software Engineering
Form of module	Lecture, exercise and tutorial
Type of module	Computer Science Fundamental
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (2+2 SWS) Self-study: 112 h per semester, including • 28 h: Preparation and studying of lectures/tutorials • 84 h: Preparation of the exams
Prerequisites	Knowledge of Java and/or Python
Aim of module	The course introduces the fundamental concepts and practices of software testing and software experimentation / evaluation • Principles of Testing • Test Coverage Criteria • Test specification techniques • Model-based testing • Principles of Experimentation • Use of Software Observatoriums
Learning outcomes and qualification goals	Professional competence: • Students will be familiar with principles, approaches and tools for testing software and the methods and empirically evaluating software products, methods and tools (MK1, MK2) Methodological competence: • Students will have the ability to judge, select and apply traditional or non-traditional test tools and techniques to evaluate software quality, as well as to design and perform software experiments (MF1, MF3) Personal competence: • Students will have improved skills in analytical thinking and applying theoretical knowledge to solve practical problems, especially in software testing and experimentation (MKO3)
Media	Lecture notes, presentations, tool demonstrations, software, data sets
Literature	Paul Ammann & Jeff Offutt., <i>Introduction to Software Testing</i> , Cambridge University Press, January 2008
Methods	Lectures, weekly exercises, experimentation with different systems
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	90 minutes

Language	English
Offering	HWS
Person in charge	Prof. Dr. Colin Atkinson
Duration of module	1 semester
Further modules	Model-Driven Development, Generative Software Engineering
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	1st/ 2nd semester

CS 550: Algorithmics

Module number	CS 550
Title	Algorithmics
Form of module	Lecture with exercise
Type of module	Fundamental in Computer Science
Level	Master
ECTS	6
Workload	Attendance: 56 h per semester (4 h per week) Self-study: 112 h per semester, including • 28 h: Preparation and reworking of lectures/tutorials • 84 h: Preparation of the exams
Prerequisites	Practical Informatics I, Algorithms and Data Structures, Linear Algebra, Statistics
Aim of module	The lecture deals with the design and the analysis of algorithms for various practically relevant computational problems and with methods for analyzing the complexity of certain problems. In particular, we will learn methods of formalizing discrete optimization problems and designing algorithms for them on the basis of analyzing the structure of these problems. Moreover, we will learn techniques for proving the correctness and estimating the running time of these algorithms. In the second part of the lecture, we will deal with the theory of NP-completeness with gives evidence that certain highly relevant problems do not have efficient algorithms. During the lecture we will derive algorithms and complexity-theoretic results for the following computational problems: • Shortest path problems and shortest round tour problems • Linear optimization problems • Flow problem • Matching problems • Satisfiability problems • Discrete linear optimization problems
Learning outcomes and qualification goals	Professional competence: • The students know efficient algorithms and the most important complexity-theoretic results for a number of computational problems which are highly relevant in practice (MK1, MK2) Methodological competence: • The students learn to formalize informally specified computational problems • They learn to analyze their structure with the goal to design efficient algorithms • They learn to prove the correctness • They learn to analyze the running time of given algorithms • They learn to prove the NP-completeness of certain problems (MF1, MF3)

	Personal competence: • Training of analytical, focussed and precise thinking • Development of abstraction abilities and the ability to transfer theoretical knowledge for solving practical problems, especially in the field of operations research • Increasing the sensitivity for the complexity and the efficient solvability of computational problems, especially through dealing with the theory of NP-completeness (MF1, MKO3)
Media	Lectures Slides and Exercises
Literature	• Cormen, Leiserson, Rivest, Stein: <i>Introduction to Algorithms</i>, 3rd edition • Shimon Even: <i>Graph Algorithms</i> • Lovasz, Plummer: <i>Matching Theory</i> • <i>Handbooks on Operations Research and Management Science</i>, Volume 7 (Editors: Ball, Magnati, Monma, Nemhauser) • J. Toran: <i>Das Erfüllbarkeitsproblem SAT</i>, Lehmann Media, 2012
Methods	Reworking of lectures and tutorials, self-studies with literature, solving exercises at home and in cooperation with other students at the tutorials
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	FSS
Person in charge	Prof. Dr. Matthias Krause
Duration of module	1 semester
Further modules	CS 651 - Cryptography II
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Mathematics in Business and Economics, M.Sc. Economic and Business Education, Teacher Education Informatics, M.Sc. Mathematics
Semester	1st/ 2nd semester

CS 600: Model-driven Development

Module number	CS 600
Title	Model-Driven Development
Form of module	Lecture with accompanying tutorials
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (2+2 SWS) Self-study: 124 h per semester, including • 32 h: Pre- and post-lecture studying and revision • 60 h: Tutorial exercises • 32 h: Examination preparation
Prerequisites	CS 500: Advanced Software Engineering
Aim of module	The course focuses on the principles, practices and tools involved in advanced model-driven development. This includes established modelling standard languages (e. g. UML, ATL, OCL) and modelling infrastructures (e. g. MOF, EMF, . . .) as well as leading edge, state-of-the-art modelling technologies (e. g. LML, PLM . . .). Key topics addressed include: • Principles of Model-Driven Development • Meta-Modeling • Domain-Specific Languages • Model Constraint and Transformation languages • Multi-View/Multi-Paradigm Modeling • Multi-Level Modeling • Ontology Modeling • Enterprise Architecture Modeling
Learning outcomes and qualification goals	Professional competence: • Students will be familiar with the accepted best practices and technologies used in mainstream model-driven development as well as state-of-the-art modeling technologies emerging from research institutions (MK1, MK2) Methodological competence: • Students will know how to apply modeling technologies in real-world projects (MF1, MF3) Personal competence: • Students will have the capability to analyse, understand and model complex systems (MKO1)
Media	Printed Lecture notes, Presentations, Tool Demonstrations
Literature	Jos B. Warmer and Anneke G. Kleppe, <i>The Object Constraint Language: Getting Your Models Ready for MDA</i> , Addison-Wesley Object Technology Series, 2003

Methods	The course consists of • Lectures that introduce, explain and exemplify the topics covered in the course • Individual projects in which each student develops a small system utilizing the topics covered in the course • Tutorials involving practical exercises and discussions of the individual projects
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr. Colin Atkinson
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	2nd/ 3rd/ 4th semester

CS 606: Foundations of Artificial Intelligence: Search and Problem Solving

Modulnummer	CS 606
Titel	Foundations of Artificial Intelligence: Search and Problem Solving
Form der Veranstaltung	Vorlesung mit Übung
Typ der Veranstaltung	Vertiefung Informatik
Modulniveau	Master
ECTS	6
Arbeitsaufwand	Präsenzstudium: 56 h pro Semester (4 SWS) Eigenstudium: ca. 98 h pro Semester, davon • 70 h: Vor- und Nachbereitung der Veranstaltung und freies Selbststudium • 28 h: Vorbereitung für die Prüfung, z.B. Prüfungs-/ Seminarabschlussarbeits- und Präsentationsvorbereitung
Vorausgesetzte Kenntnisse	Formale Grundlagen der Informatik (insbes. Komplexität, Graphen und Logik), Algorithmen & Datenstrukturen, Programmierpraktikum I & II
Lehrinhalte	• Problemeigenschaften und Problemtypen • Problemlösen als Suche, Anwendung im Bereich Spiele • Constraintprobleme und deren Lösung • Logische Constraints • Entscheidungsbaumlernen
Lern- und Kompetenzziele	Fachkompetenz: • Ziele und Grundlagen der Künstlichen Intelligenz • Suchverfahren als universelle Problemlösungsverfahren • Problemkomplexität und Heuristische Lösungen • Eigenschaften und Zusammenhang zwischen unterschiedlichen Suchverfahren (BK7) Methodenkompetenz: • Beschreibung konkreter Aufgaben als Such-, Constraint- oder Planungsproblem • Implementierung unterschiedlicher Suchverfahren und Heuristiken (BF1, BKO1) Personale Kompetenzen: -
Medienformen	Lehrbuch, Vorlesung, Präsentationen, Tafel
Begleitende Literatur	Russel and Norvig: <i>Artificial Intelligence - a modern approach</i> . Prentice Hall. 3rd Edition 2010
Lehr- und Lernmethoden	Vorlesung, praktische Übungen, eigenständige Bearbeitung von Übungsaufgaben
Art der Prüfungsleistung	Schriftliche Prüfung
Prüfungsvorleistung	• 1-2 Mini-Projekte erfolgreich bearbeitet und dokumentiert • Mindestens 60% aller Aufgabenpunkte aufsummiert

Prüfungsdauer	90 Minuten
Sprache	Deutsch/Englisch
Angebotsturnus	HWS
Modulverantwortliche/r	Prof. Dr. Heiner Stuckenschmidt
Dauer des Moduls	1 Semester
Weiterführende Module	SM 442 Bachelorseminar Prof. Stuckenschmidt
Verwendbarkeit	M.Sc. Wirtschaftsinformatik, B.Sc. Wirtschaftsinformatik, M.Sc. Wirtschaftspädagogik, Lehramt Informatik, Beifach Angewandte Informatik, M.Sc. Data Science
Einordnung in Fachsemester	5./ 6. Fachsemester im Bachelor, 1./ 2./ 3. Fachsemester im Master

CS 651: Cryptography II

Module number	CS 651
Title	Cryptography II
Form of module	Inverted classroom
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 112 h per semester
Prerequisites	Even though the lecture deepens and continues topics discussed in “CS 404 Cryptography I”, it is not a prerequisite to have attended this lecture. The lecture “Cryptography II” can be studied without any prior knowledge on cryptography - all necessary basics are shortly recapitulated
Aim of module	The goal of this lecture is to present and discuss important scientific concepts from modern cryptography. Possible topics are: • Security Definitions: How can the security of cryptographic schemes formally defined? • Proofs of Security: How can the security of cryptographic schemes be proven (based on precise assumptions)? • Cryptanalysis: What are the established techniques to analyze cryptographic mechanisms? • Elliptic Curves • Zero-Knowledge Proofs
Learning outcomes and qualification goals	Professional competence: • With the help of current techniques and theories of modern cryptography, the students can assess the security of cryptographic processes and assess security statements accordingly • They can identify security goals and use appropriate techniques that could not be dealt with in “CS 404 Cryptography I” (MK2) Methodological competence: • The students can select and use suitable methods for the security analysis of cryptographic processes. This includes, for example, the choice of the appropriate security model, proof of security based on clearly specified assumptions and the analysis of given procedures • The students can understand and assess the security arguments for existing procedures and to transfer them to new ones. • They can use techniques and protocols to achieve security goals that were not yet possible with the methods discussed in “CS 404 Cryptography I” (MK1)

	Personal competence: • The analytic, concentrated and precise thinking of the students is trained • By the independent treatment of applications, e.g. in the course of the exercises, their abstraction capacity is further developed and the transfer of the learned material to related questions is trained (MF1, MKO3)
Media	Video recordings, annotated lecture slides
Literature	Jonathan Katz, Yehuda Lindell: <i>Introduction to Modern Cryptography: Principles and Protocols</i> , Chapman and Hall/CRC, 2007
Methods	Reworking the lecture and studying the relevant literature in self-study. During the lecture: discussing questions and ideas and working together on concrete examples. Solving exercises in self-study and in practice in cooperation with fellow students
Form of assessment	Oral examination
Admission requirements for assessment	-
Duration of assessment	30 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr. Frederik Armknecht
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science, M.Sc. Business Informatics, Teacher Education Informatics, M.Sc. Mathematics
Semester	1st/ 2nd/ 3rd semester

CS 664: Blockchain Security

Module number	CS 664
Title	Blockchain Security
Form of module	Inverted classroom with exercises
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 112 h per semester
Prerequisites	There are no formal prerequisites but knowledge in cryptography and/or IT-security is recommended, e.g., by attending the lectures “Kryptographie I” or “Selected Topics in IT-Security”
Aim of module	Besides cryptocurrencies, blockchain technology has become increasingly important for a variety of use cases such as finance, supply chain management, digital identity, and decentralized applications. The main difference to previous technologies is that it promises secure and reliable data storage and consensus in a trustless environment. Still, we observe a lack of understanding on the security properties and limitations of blockchain technology. The aim of this lecture is to close this gap by focusing on the security of blockchain technology. The course will equip students with a solid understanding of blockchains, their design principles, underlying technologies, and cryptographic primitives. Bitcoin, Monero and Ethereum will be discussed in greater detail and a substantial part of the course will be devoted to security issues and possible attacks
Learning outcomes and qualification goals	Professional competence: • Students will acquire profound knowledge of Blockchain technology as well as the skills to critically examine the security of Blockchain-based systems (MK1, MK2) Methodological competence: • Successful participants will be able to understand and evaluate the different ways in which different Blockchain systems try to achieve security • They will also be able to identify where, why and how these security measures are broken for both, current and new systems (MKO3) Personal competence: • The analytic, concentrated and precise thinking of the students is trained • As multiple different but related Blockchains are discussed, their abstraction capacity is further developed and the transfer of the learned concepts to related questions is trained (MF1)
Media	Video recordings, annotated lecture slides
Literature	Will be announced in the lecture

Methods	Reworking the lecture and studying the relevant literature in self-study. During the lecture: discussing questions and ideas and working together on concrete examples. Solving exercises in self-study and in practice in cooperation with fellow students
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr. Frederik Armknecht
Duration of Module	1 Semester
Further Modules	-
Programs	M.Sc. Data Science, M.Sc. Business Informatics, Teacher Education Informatics, M.Sc. Mathematics, M.Sc. Mathematics in Business and Economics
Semester	1st/ 2nd/ 3rd semester

IE 650: Knowledge Graphs

Module number	IE 650
Title	Knowledge Graphs
Form of module	Lecture
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 124 h per semester, including • 82 h: Pre- and post-lecture studying and revision • 42 h: Examination preparation
Prerequisites	Java or Python programming skills
Aim of module	• The Role of knowledge graphs in the AI landscape • Semantic Web and its representation languages • Labeled property graphs • Query languages for knowledge graphs • Knowledge modeling and ontologies • Logical reasoning with knowledge graphs • Machine learning with knowledge graphs and knowledge graph embeddings
Learning outcomes and qualification goals	Professional competence: • The participants of this course learn about principles and applications of knowledge graphs • They become familiar with their technical foundations such as representation and query languages, or logical inference • After taking this course, the students will be aware of the problems and benefits of knowledge graph technologies in the context of tasks such as knowledge management, information search and data integration • They will be capable of judging the applicability of these technologies for addressing practical challenges (MK1, MK2) Methodological competence: • The participants learn how to design and implement AI systems based on knowledge graphs • They are able to use standardized modeling languages for building knowledge representations and to query these models by means of languages such as SPARQL (MF3) Personal competence: • By jointly building a knowledge graph-based application, the students learn how to effectively work in teams • They improve upon their presentation skills by showing the outcomes of their projects to the other participants of the course (MKO1, MKO3)
Media	Lecture slides and exercise sheets will be available online

Literature	• Pascal Hitzler, Markus Krötzsch and Sebastian Rudolph, <i>Foundations of Semantic Web Technologies</i>, Chapman & Hall/CRC, 2009 • Allemang and Hendler (2008): <i>Semantic Web for the Working Ontologist</i>. Verlag Morgan Kaufmann • Antoniou and van Harmelen (2004): <i>A Semantic Web Primer</i>. MIT Press • Fensel et al. (2020): <i>Knowledge Graphs: Methodology, Tools and Selected Use Cases</i>. Springer • Kerjwal et al. (2021): <i>Knowledge Graphs: Fundamentals, Techniques, and Applications</i>. MIT Press
Methods	The course participants will take part in theoretical and practical exercises, the solutions of which are discussed in the tutorials. At the end of the course, they get the opportunity to apply their knowledge in a team project. Each student team will design and implement a semantic web application, and subsequently present the results to the other students. Besides the exercises, regular presentations including references to relevant course materials and recommended readings will be given by the lecturer. The lecturer as well as the tutors offer individual help and consulting to the participants of the course
Form of assessment	Written examination
Admission requirements for assessment	Project report and oral presentation
Duration of assessment	60 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr. Heiko Paulheim, Dr. Sven Hertling
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Social Data Science, Teacher Education Informatics
Semester	1st/ 2nd/ 3rd semester

IE 670: Web Data Integration

Module number	IE 670
Title	Web Data Integration
Form of module	Lecture
Type of module	Specialization Course
Level	Master
ECTS	3
Workload	Hours in presence: 28 h per semester (2 SWS) Self-study: 56 h per semester, including • 31 h: Pre- and post-lecture studying and revision • 25 h: Examination preparation
Prerequisites	-
Aim of module	Data integration is one of the key challenges in most IT projects and it is estimated that data scientists spend about 80% of their time on data integration and data preparation. Within the enterprise context, data integration problems arise whenever data from separate sources needs to be combined as the basis for new applications or data analysis projects. Within the context of the Web, data integration techniques form the foundation for taking advantage of the ever-growing number of publicly-accessible data sources. The course will cover the following topics: • Heterogeneity and Distributedness • The Data Integration Process • Structured Data on the Web • Data Exchange Formats • Schema Mapping and Data Translation • Identity Resolution • Data Quality Assessment • Data Fusion It is highly recommended to attend the course IE683 Web Data Integration Project in the same semester as this course as the schedules of both courses are aligned to each other
Learning outcomes and qualification goals	Professional competence: • Students will be able to identify opportunities for employing Web data in business applications and will learn to select and apply appropriate techniques for integrating and cleansing Web data (MK1, MF1) Methodological competence: • Participants will acquire knowledge of the data integration process as well as the techniques that are used in each phase of the process (MK2, MF3, MF4, MKO3) Personal competence: -
Media	Slide set

Literature	• AnHai Doan, Alon Halevy, Zachary Ives: <i>Principles of Data Integration</i>. Morgan Kaufmann, 2012 • Luna Dong, Divesh Srivastava: <i>Big Data Integration</i>. Morgan & Claypool, 2015 • Ulf Leser, Felix Naumann: <i>Informationsintegration</i>. Dpunkt Verlag, 2007
Methods	The course consists of a lecture that introduces students to state of the art data integration techniques
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	60 minutes
Language	English
Offering	HWS
Person in charge	Dr. Ralph Peeters
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Social Data Science, Teacher Education Informatics
Semester	1st/ 2nd/ 3rd semester

IE 683: Web Data Integration Project

Module number	IE 683
Title	Web Data Integration Project
Form of module	Project
Type of module	Specialization Course
Level	Master
ECTS	3
Workload	Hours in presence: 28 h per semester (2 SWS) Self-study: 56 h per semester, including • 36 h: Project work • 20 h: Report writing and presentation preparation
Prerequisites	Programming skills in Python
Aim of module	The web data integration project allows students to apply the methods and techniques that they have learned in the lecture Web Data Integration in the context of a practical integration project. The projects cover all steps of the data integration process including data gathering, schema mapping, data translation, identity resolution, data quality assessment, and data fusion. It is highly recommended to attend the web data integration lecture (IE670) in the same semester as the web data integration project as the schedules of both courses are aligned to each other
Learning outcomes and qualification goals	Professional competence: • Students will be able to identify opportunities for employing Web data in business applications and will learn to apply appropriate techniques for integrating and cleansing Web data (MK1, MF1) Methodological competence: • Participants will acquire knowledge of the data integration process as well as the techniques that are used in each phase of the process • Project organization skills (MK2, MF3, MF4, MKO3) Personal competence: • Presentation skills • Team work skills (MKO2, MF2)
Media	Exercise sheets, Java project template
Literature	• AnHai Doan, Alon Halevy, Zachary Ives: <i>Principles of Data Integration</i>. Morgan Kaufmann, 2012 • Luna Dong, Divesh Srivastava: <i>Big Data Integration</i>. Morgan & Claypool, 2015 • Ulf Leser, Felix Naumann: <i>Informationsintegration</i>. Dpunkt Verlag, 2007
Methods	Students work on their integration projects in teams and will report about the results of their projects in the form of a written report as well as an oral presentation

Form of assessment	• Project report (70%, 10-20 pages in length) • Oral project presentation (30%, 15-20 minutes)
Admission requirements for assessment	-
Duration of assessment	-
Language	English
Offering	HWS
Person in charge	Dr. Ralph Peeters
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Social Data Science, Teacher Education Informatics
Semester	1st/ 2nd/ 3rd semester

DS 200: Sampling and Data

Module number	DS 200
Title	Sampling and Data
Form of module	Lecture and Tutorial
Type of module	Data Science Methods: Fundamentals
Level	Master
ECTS	9
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 214 h per semester, including • 91 h: Pre- and post-lecture studying and revision • 42 h: Studying for and taking weekly online tests • 41 h: Preparation and presentation of weekly exercises • 40 h: Examination preparation
Prerequisites	-
Aim of module	The course introduces the relevant methods for research design and data analysis in the social science. The course will center on causal inference, identification, and estimation. Reading and critiquing papers will be emphasized along with the data and statistical analyses
Learning outcomes and qualification goals	Professional competence: • Students can independently formulate research questions and empirically verifiable hypotheses (MK1) Methodological competence: • Students know different data collection methods and their advantages and disadvantages • They are able to assess the appropriateness of the methods used to answer various questions • They understand and critically evaluate the design of empirical studies (MK1) Personal competence: • The course supports students to develop competences with regard to choosing the appropriate research design and sampling frame (MF2, MKO1, MKO2)
Media	Lecture slides available online
Literature	• Angrist, Joshua D., and Jörn-Steffen Pischke. <i>Mostly Harmless Econometrics: An Empiricist's Companion</i>. Princeton: Princeton University Press, 2009 • Imbens, Guido W., and Donald B. Rubin. <i>Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction</i>. Cambridge: Cambridge University Press, 2015 • Cunningham, Scott. <i>Causal Inference: The Mixtape</i>. New Haven and London: Yale University Press, 2021
Methods	Lecture elements, weekly tests, literature studies
Form of assessment	Written examination

Admission requirements for assessment	Oral participation, homework, presentations, compulsory attendance
Duration of assessment	90 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr. Marc Ratkovic
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science, M.Sc. Social Data Science
Semester	-

D. Data Analytics Methods

1. Overview

Module no.	Name of Module	Offered	Language	ECTS	Page
AC 652	Additional Course - Data Analytics Methods	HWS/FSS	E	Max. 18	37
DS 100	Statistics for Data Scientists	HWS	E	9	38
DS 201	Machine Learning and Causal Inference	FSS	E	9	40
DS 202	Seminar and Lab on Machine Learning and Causal Inference	HWS	E	9	42
DS 204	Mathematical Foundations of machine Learning for Social Data Scientists	FSS	E	6	44
IE 560	Foundations of Artificial Intelligence - Reasoning and Decision Making	HWS	E	6	46
IE 678	Deep Learning	FSS	E	6	48
IE 679	AI Deployment and Operations	HWS	E	3	50
IE 685	Large Language Models and Agents	FSS	E	3	52
IE 686	Large Language Models and Agents Project	FSS	E	3	54
IE 692	Advanced Process Mining	FSS	E	6	56
IE 695	Reinforcement Learning	HWS	E	6	58
IE 696	Advanced Methods in Text Analytics	FSS	E	6	60
IS 617 (IS 6170)	Large Language Models for the Economic and Social Sciences	HWS	E	6	MMM*
IS 661 (IS 6610)	Text Analytics	HWS	E	6	MMM*
MAA 519	Stochastic Calculus	FSS	E	5	MBE*
MAB 508	Algebraische Statistik	irregular	G/E	8	MBE*
MAB 519	Reinforcement Learning	FSS	E	10	MBE*
MAB 520	Reinforcement Learning II	HWS	E	5	MBE*
MAC 404	Optimierung	HWS	G	8	WM*
MAC 507	Nonlinear Optimization	FSS	E	6	MBE*
CS 646	Higher Level Computer Vision	HWS	E	6	62
CS 647	Image Processing	FSS	E	6	64
CS 668	Generative Computer Vision Models	FSS	E	6	66

* For a detailed description, please see the module catalogues of the respective following degree programs:

- WM: B.Sc. Wirtschaftsmathematik (only available in German)
<https://www.wim.uni-mannheim.de/studium/studienorganisation/b-sc-wirtschaftsmathematik/>
- MBE: M.Sc. Mathematics in Business and Economics (only available in German)
<https://www.wim.uni-mannheim.de/studium/studienorganisation/m-sc-wirtschaftsmathematik/>
- MMM: Mannheim Master in Management
<https://www.bwl.uni-mannheim.de/en/module-catalog-mmm/>

2. Detailed descriptions

AC 652: Additional Course - Data Analytics Methods

Module number	AC 652
Title	Additional Course - Data Analytics Methods
Form of module	Depends on course
Type of module	Depends on course
Level	Master
ECTS	Max. 18(A total of 18 ECTS can be recognized over all possible additional courses)
Workload	Depends on course
Prerequisites	Depends on course
Aim of module	The course falls into the data analytics methods area of the MMDS and covers topics related to data analytics methods but is not directly equivalent to any course in the MMDS module catalogue. The course level equals a regular course in MMDS study program. The module can be taken either at the University of Mannheim or at any other university in Germany or abroad
Learning outcomes and qualification goals	Depends on course
Media	Depends on course
Literature	Depends on course
Methods	Depends on course
Form of assessment	Depends on course
Admission requirements for assessment	Depends on course
Duration of assessment	Depends on course
Language	English preferred, but any other language possible if Mannheim faculty member is able to identify content and level
Offering	HWS/FSS
Person in charge	Lecturer at the host university
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science
Semester	2nd/ 3rd/ 4th semester

DS 100: Statistics for Data Scientists

Module number	DS 100
Title	Statistics for Data Scientists
Form of module	Lecture and Tutorial
Type of module	Foundations of Data Science
Level	Master
ECTS	9
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 214 h per semester, including • 91 h: Pre- and post-lecture/tutorial studying and revision • 42 h: Studying for and taking weekly online tests • 40 h: Examination preparation • 41 h: Preparation and presentation of weekly exercises
Prerequisites	A sound introduction of the linear regression model (OLS) is required. Knowledge in linear algebra and calculus is useful. This is the first in a two-course sequence covering statistical models and causal inference, continuing into Data Science 201 next semester. The topics in both courses will be integrated through the year. It is expected that students will take both semesters
Aim of module	The course provides an introduction to the basics of regression models, including: • The theoretical development of linear regression • Extensions to generalized linear models • Extensions to machine learning and model selection • Regular practice on applied data analysis
Learning outcomes and qualification goals	Professional competence: • Understand how to appropriately translate research question into statistical models • Be able to apply statistical models appropriate for non-linear problems • Learn how to present and interpret estimation results in a substantive meaningful way (MK1, MK3) Methodological competence: • Estimate regression parameters using the maximum likelihood principle • Perform hypothesis tests for regression models using the maximum likelihood principle • Be able to identify violations of the respective regression assumptions of the discussed GLMs • Be able to identify limitations of non-linear regression models (MK1, MK3)

	Personal competence: • The course supports students to develop competences with regard to choosing the appropriate statistical method(s) to answer respective research questions and how to present and communicate statistical results (MF1, MF2, MF3, MKO1, MKO2)
Media	Lecture slides available online, exercises available online
Literature	• Verbeek, M. 2017. <i>A Guide to Modern Econometrics</i>. 5th ed. Chichester: Wiley • Hansen, Bruce E. 2022. <i>Econometrics</i>. Princeton: Princeton University Press • Angrist, Joshua D., and Jörn-Steffen Pischke. 2009. <i>Mostly Harmless Econometrics: An Empiricist's Companion</i>. Princeton: Princeton University Press • Wooldridge, J.M. 2002. <i>Econometric Analysis of Cross Section and Panel Data</i>. Cambridge, MA: MIT Press
Methods	Lecture elements, weekly tests, literature studies
Form of assessment	Written examination
Admission requirements for assessment	Oral participation, homework, presentations, compulsory attendance
Duration of assessment	90 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr. Marc Ratkovic
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Social Data Science, M.Sc. Data Science
Semester	1st semester

DS 201: Machine Learning and Causal Inference

Module number	DS 201
Title	Machine Learning and Causal Inference
Form of module	Lecture and Tutorial
Type of module	Data Science Methods: Fundamentals
Level	Master
ECTS	9
Workload	Hours in presence: 56 h per semester (2 SWS) Self-study: 214 h per semester, including • 91 h: Pre- and post-lecture studying and revision • 42 h: Studying for and taking weekly online tests • 41 h: Preparation and presentation of weekly exercises • 40 h: Examination preparation
Prerequisites	A sound understanding of basic statistical methods incl. regression analysis, estimation methods (OLS, MLE) and statistical inference is required. Knowledge in linear algebra and calculus is also required. This is the third in a three-course sequence covering statistical models and causal inference for MMSDS. The topics in all three courses (DS 100, DS 200) will be integrated. It is expected that students will take both semesters
Aim of module	The course aims to introduce students to likelihood-based and Bayesian methods, machine learning methods, then deep learning methods. Extensions and connections to causal methods will be made at each stage. Advanced and popular techniques of machine learning are introduced
Learning outcomes and qualification goals	Professional competence: • Students are aware of how machine learning can help answer causal questions • They can analyze (large) data sets using popular techniques of machine learning (MK1) Methodological competence: • Students are able to translate their research question(s) into a causal framework • They can apply different techniques of machine learning (MK1, MK3) Personal competence: • The course supports students to develop competences with regard to choosing the appropriate statistical method(s) to answer respective research questions (MF1, MF2, MF4, MKO1, MKO2)
Media	Lecture slides available online

Literature	• McElreath, Richard. 2020. <i>Statistical Rethinking: A Bayesian Course with Examples in R and STAN</i>. Routledge • Murphy, Kevin P. 2012. <i>Machine Learning: A Probabilistic Perspective</i>. Cambridge: MIT Press • Hastie, Trevor, Robert Tibshirani, and Jerome Friedman. 2008. <i>The Elements of Statistical Learning</i>. 2nd ed. New York: Springer • Jurafsky, Daniel, and James H. Martin. 2009. <i>Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition</i>. 2nd ed. Upper Saddle River, NJ: Prentice Hall • Imbens, Guido W., and Donald B. Rubin. 2015. <i>Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction</i>. Cambridge: Cambridge University Press
Methods	Lecture elements, weekly tests, literature studies
Form of assessment	Written examination
Admission requirements for assessment	Oral participation, homework, presentations, compulsory attendance
Duration of assessment	90 minutes
Language	English
Offering	FSS
Person in charge	Prof. Dr. Marc Ratkovic
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Social Data Science, M.Sc. Data Science
Semester	2nd semester

DS 202: Seminar and Lab on Machine Learning and Causal Inference

Module number	DS 202
Title	Seminar and Lab on Machine Learning and Causal Inference
Form of module	Seminar
Type of module	Data Science Methods: Fundamentals
Level	Master
ECTS	9
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 214 h per semester, including • 70 h: Pre- and post-seminar studying and revision • 70 h: Analyzing data, programming Python scripts • 74 h: Preparation and presentation of research project
Prerequisites	Students should be familiar with Python or R or at least with any other object-programming language
Aim of module	The module introduces, discusses and offers hands-on experience in state-of-the-art topics in data science. It consists of a “theoretical” part (seminar) which offers room for discussion on hot topics of data science and an “empirical” part where students formalize research questions and analyze data (of their own) using advanced techniques of machine learning
Learning outcomes and qualification goals	Professional competence: • Students are able to find appropriate techniques of machine learning to answer complex research questions • They can analyze (large) data sets using state-of-the-art techniques of machine learning (MK1) Methodological competence: • Students are able to program in Python • They can apply advanced techniques of machine learning to individual research questions • They have in-depth knowledge in application of data science (MK1, MK2, MK3) Personal competence: • Students are able to independently implement their research question in a research project • They can deal with and solve theoretical and empirical problems (MF1, MF2, MF3, MF4, MKO1, MKO2)
Media	Slides available online

Literature	• Downey, Allen B. 2016. <i>Think Python. How to Think Like a Computer Scientist</i>. 2nd ed. Needham: Green Tea Press. (available online: http://greenteapress.com/thinkpython2/thinkpython2.pdf) • McElreath, Richard. 2020. <i>Statistical Rethinking: A Bayesian Course with Examples in R and STAN</i>. Routledge • Martin and Jurafsky. 2024. <i>Speech and Language Processing</i>. Available at: https://web.stanford.edu/~jurafsky/slp3/ • Guttag, John V. 2021. <i>Introduction to Computation and Programming Using Python</i>. 3rd ed. Cambridge: MIT Press • The Python Tutorial (online: https://docs.python.org/3/tutorial/)
Methods	Lecture elements, classroom discussion, weekly lab-sessions, literature studies, external speakers
Form of assessment	Written examination
Admission requirements for assessment	Oral participation, homework, presentations, compulsory attendance
Duration of assessment	90 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr. Marc Ratkovic
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Social Data Science
Semester	3rd semester

DS 204: Mathematical Foundations of machine Learning for Social Data Scientists

Module number	DS 204
Title	Mathematical Foundations of Machine Learning for Social Data Scientists
Form of module	Lecture with tutorial
Type of module	Data Analytics Methods
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Exercises and self-study: 98 h per semester
Prerequisites	• This is an introductory course and has no pre-requisites • Data Science 100 or 200 is recommended
Aim of module	The aim of the module is to equip students with the mathematics necessary to understand and work with machine learning models. This is an introductory course, pre-supposing only first year college level mathematics, at the level of basic algebra, calculus and geometry. The module is aimed to get students from their current mathematical location to one where they can understand the basics of mathematics for machine learning, as well as allow them to study more advanced topics with a mathematical dimension. We will cover the material in the first seven chapters of <i>Mathematics for Machine Learning</i>, with a focus on depth of understanding rather than covering a broad range of topics
Learning outcomes and qualification goals	Professional competence: • The student will gain proficiency in linear algebra, probability theory and optimization Methodological competence: • The student will be able to combine the methods learned in the class with models encountered in applied questions and work in the social sciences Personal competence: • The student will learn basic proof structures and argument structures that can be used to strengthen their theses and professional arguments
Media	Slides will be prepared along with lecture notes; students will collaborate on a set of shared solutions
Literature	<i>Mathematics for Machine Learning</i> (2020). Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong. Cambridge University Press
Methods	The course will involve a lecture where material is discussed as well as a tutorial where material is practiced
Form of assessment	Written examination

Admission requirements for assessment	There will be assessments through the semester that will need to be passed in order to take the exam. These may include assessments in lecture or tutorial, as well as take-home assignments
Duration of assessment	90 minutes
Language	English
Offering	FSS
Person in charge	Prof. Dr. Marc Ratkovic
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science, M.Sc. Social Data Science
Semester	1st/ 2nd semester

IE 560: Foundations of Artificial Intelligence - Reasoning and Decision Making

Module number	IE 560
Title	Foundations of Artificial Intelligence - Reasoning and Decision Making
Form of module	Inverted classroom
Type of module	Business Informatics Fundamental
Level	Master
ECTS	6
Workload	Hours in presence: 50 h per semester (2+2 SWS) Self-study: 120 h per semester, including • 80 h: Pre- and post-lecture studying and revision • 40 h: Examination preparation
Prerequisites	Linear Algebra, Basic Probability Theory, Basic Knowledge of Propositional Logic
Aim of module	The course provides an introduction to methods for AI-based decision making as a basis for the design of automated decision making and decision support systems. The course will cover the following topics: • Decision Problems and Bounded Rationality • Propositional Logic • Probability Theory • Probabilistic Graphical Models • Utility and Decision Theory • Game Theory • Markov Decision Processes • Reinforcement Learning
Learning outcomes and qualification goals	Professional competence: • Students will acquire basic knowledge of the techniques, opportunities and applications of AI-based decision making (MK1, MF1) Methodological competence: • Successful participants will be able to identify opportunities for decision support in an enterprise environment, select and apply appropriate techniques, and interpret the results (MK2, MF3, MF4, MKO1) Personal competence: -
Media	Lecture videos, slide set, exercise sheets, software tools
Literature	S. Russel and P. Norvig: <i>AI: A Modern Approach</i> (3rd Edition), 2010 (selected sections)
Methods	The course consists of a lecture accompanied by practical homework and case studies. In the lecture, the students' basic concepts and methods of decision theory will be explained both in theory and using concrete examples. The students will practice and test their knowledge acquired in the lecture in homework assignments. Within the case studies, students model real world decision problems and try to solve them optimally using methods from the lecture

Form of assessment	Written examination
Admission requirements for assessment	Midterm exam: 45 minutes (propositional logic and probability theory)
Duration of assessment	90 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr. Heiner Stuckenschmidt
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Social Data Science
Semester	1st/ 2nd semester

IE 678: Deep Learning

Module number	IE 678
Title	Deep Learning
Form of module	Lecture with exercise
Type of module	Data analytics methods (MMDS), Specialization Course (M.Sc. Wifo)
Level	Master
ECTS	6
Workload	Hours in presence: 42 h per semester (2+2 SWS) Exercises, assignments and self-study: 119 h per semester
Prerequisites	IE 675b Machine Learning or equivalent
Aim of module	Building upon IE 675b Machine Learning, this course focuses on deep learning and introduces basic and advanced deep learning concepts, architectures, training methods as well as selected applications. Topics include: • Feedforward neural networks • Backpropagation, optimizers and architecture design • Machine learning systems • Local and global embeddings • Convolutional neural networks • Recurrent neural networks and structured state space models • Attention and Transformers • Deep learning for graphs • Training techniques for deep learning models, including data augmentation, pretraining and fine-tuning and in-context learning
Learning outcomes and qualification goals	Professional competence: • Deep understanding of fundamental concepts, models and algorithms of deep learning (MK1, MF1) Methodological competence: • Being able to build and train deep learning models • Being able to select suitable deep learning techniques for a given learning problem (MK2, MF3, MF4) Personal competence: • Writing skills • Presentation skills • Statistical programming skills (MKO3, MF2)
Media	Slide set, exercise sheets, software, datasets

Literature	• K.P. Murphy. <i>Probabilistic Machine Learning: An Introduction</i>. The MIT Press, 2022 • K.P. Murphy. <i>Probabilistic Machine Learning: Advanced Topics</i>. The MIT Press, 2023 • I. Goodfellow, Y. Bengio, A. Courville. <i>Deep Learning</i>. The MIT Press, 2017 • I. Drori. <i>The Science of Deep Learning</i>. Cambridge University Press, 2023 • Additional material and articles provided in lecture notes
Methods	The course consists of a lecture accompanied by theoretical and practical exercises as well as case studies with real data. In the exercises, students will deepen the material discussed in the lecture, apply the methods in practice, and present the result
Form of assessment	Oral examination
Admission requirements for assessment	Homework assignments (pass at least 2 assignments)
Duration of assessment	25 minutes
Language	English
Offering	FSS
Person in charge	Prof. Dr. Rainer Gemulla
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	2nd/ 3rd semester

IE 679: AI Deployment and Operations

Modulnummer	IE 679
Titel	AI Deployment and Operations
Form der Veranstaltung	Online-Vorlesung + praktische Gruppenarbeit
Typ der Veranstaltung	Spezialisierung / Data Analysis Methods
Modulniveau	Master
ECTS	3
Arbeitsaufwand	Präsenzstudium: 14 h pro Semester (1 SWS) Eigenstudium: 75 h pro Semester, davon • 60 h: Vor- und Nachbereitung der Veranstaltung und freies Selbststudium • 15 h: Vorbereitung für die Prüfung
Vorausgesetzte Kenntnisse	Machine Learning oder Data Mining I
Lehrinhalte	• Grundsätze von Cloudcomputing und praktische Anwendung • Infrastructure as Code • Gängige Infrastrukturen • Erstellung von replizierbarem Code nach Industriestandards • Erstellung von Dashboards zur Analyse von Daten
Lern- und Kompetenzziele	Fachkompetenz: • Unterschiede zwischen akademischen und industriellen KI-Anwendungen • Kenntnisse über den Infrastruktur-Stack für industrielle KI-Anwendungen • Voraussetzungen und Methoden für die Realisierbarkeit von Code Methodenkompetenz: • Erstellung von praxistauglichen KI Anwendungen Personale Kompetenz: • Besseres Verständnis der Anforderungen an nicht akademische Tätigkeiten
Medienformen	Präsentation
Begleitende Literatur	• Cloud Computing Buch: Learning Microsoft Azure: <i>Cloud Computing and Development Fundamentals</i> von Jonah Carrio Andersson • Dokumentation MS Azure: https://learn.microsoft.com/en-us/training/azure/ • Dokumentation Terraform: https://developer.hashicorp.com/terraform/intro
Lehr- und Lernmethoden	• Vorlesung (2 SWS) jede 2. Woche • Praktische Arbeit in Kleingruppen
Art der Prüfungsleistung	Abgabe eines Codingprojekts + Gruppenpräsentation

Prüfungsvorleistung	Dokumentation der praktischen Arbeiten
Prüfungsdauer	10-15 Minuten pro Teilnehmer
Sprache	Englisch
Angebotsturnus	HWS
Modulverantwortliche/r	Prof. Dr. Heiner Stuckenschmidt
Dauer des Moduls	1 Semester
Weiterführende Module	-
Verwendbarkeit	M.Sc. Wirtschaftsinformatik, M.Sc. Data Science, M.Sc. Social Data Science
Einordnung in Fachsemester	2./ 3./ 4. Fachsemester

IE 685: Large Language Models and Agents

Module number	IE 685
Title	Large Language Models and Agents
Form of module	Lecture
Type of module	Specialization Course
Level	Master
ECTS	3
Workload	Hours in presence: 28 h per semester (2 SWS) Self-study: 56 h per semester, including • 31 h: Pre and post lecture studying and revision • 25 h: Examination preparation
Prerequisites	• Machine learning concepts and techniques • Knowledge about natural language processing
Aim of module	Large Language Models (LLMs) such as GPT, Llama, Gemini and Claude, have the potential to enable a wide range of new applications and to significantly improve the performance of existing systems. The course introduces students to LLMs as well as their application. The course covers the following topics: • Introduction to LLMs • Prompt engineering • LLM-based agents • Tool use and environment interaction • Retrieval augmented generation • Context engineering for LLM agents • Safety and security of LLM agents • Evaluation of LLM agents It is highly recommended to attend the course IE 686 Large Language Models and Agents Project in the same semester as this course as the schedules of both courses are aligned to each other
Learning outcomes and qualification goals	Professional competence: • Students will be able to identify opportunities for employing Large Language Models in business applications and will learn to select and apply appropriate prompt/context engineering techniques for solving complex tasks (MK1, MF1) Methodological competence: • Participants will acquire knowledge on the application of Large Language Models and LLM agents for solving real-world problems (MK2, MF3, MF4, MKO3) Personal competence: -
Media	Slide sets

Literature	• Zhao, et al.: <i>A Survey of Large Language Models</i>. arXiv:2303.18223v16, 2025 • Wang, et al.: <i>A Survey on Large Language Model based Autonomous Agents</i>. Frontiers of Computer Science, 2024 • Mohammadi, et al. <i>Evaluation and benchmarking of LLM agents: A survey</i>. SIGKDD Conference on Knowledge Discovery and Data Mining, 2025
Methods	Methods The course consists of a lecture that introduces students to Large Language Models and state of the art prompt/context engineering techniques
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	60 minutes
Language	English
Offering	FSS
Person in charge	Prof. Dr. Christian Bizer
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science, M.Sc. Social Data Science, M.Sc. Business Informatics, Teacher Education Informatics
Semester	2nd/ 3rd semester

IE 686: Large Language Models and Agents Project

Module number	IE 686
Title	Large Language Models and Agents Project
Form of module	Project
Type of module	Specialization Course
Level	Master
ECTS	3
Workload	Hours in presence: 28 h per semester (2 SWS) Self-study: 56 h per semester, including • 36 h: Project work • 20 h: Report writing and presentation preparation
Prerequisites	• Programming skills in Python • Machine learning concepts and techniques
Aim of module	The Large Language Models and Agents project allows students to apply the methods and techniques that they have learned in the lecture Large Language Models and Agents (IE 685) in the context of a practical project. The projects cover all steps of the agent development cycle including agentic workflow design, prompt- and context engineering, and agent evaluation. It is highly recommended to attend the Large Language Models and Agents lecture (IE 685) in the same semester as the Large Language Models and Agents project
Learning outcomes and qualification goals	Professional competence: • Students will be able to identify opportunities for employing large language models and agents in business applications and will learn to select and apply appropriate prompt/context engineering techniques for solving complex tasks (MK1, MF1) Methodological competence: • Participants will acquire knowledge on the application of large language models and related techniques for solving complex real-world problems • Agent development using LangChain framework • Project organization skills (MK2, MF3, MF4, MKO3) Personal competence: • Presentation skills • Teamwork skills • Python programming skills (MKO2, MF2)
Media	Exercise sheets, software, datasets

Literature	• Zhao, et al.: <i>A Survey of Large Language Models</i>. arXiv:2303.18223v16, 2025 • Wang, et al.: <i>A Survey on Large Language Model based Autonomous Agents</i>. Frontiers of Computer Science, 2024 • Mohammadi, et al. <i>Evaluation and benchmarking of LLM agents: A survey</i>. SIGKDD Conference on Knowledge Discovery and Data Mining, 2025
Methods	Students work on their projects in teams and will report about the results of their projects in the form of a written report as well as an oral presentation
Form of assessment	• Project report (50%, 10-20 pages in length) • Oral project presentation and Q&A (50%, 15-20 minutes)
Admission requirements for assessment	-
Duration of assessment	-
Language	English
Offering	FSS
Person in charge	Prof. Dr. Christian Bizer
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science, M.Sc. Social Data Science, M.Sc. Business Informatics, Teacher Education Informatics
Semester	2nd/ 3rd semester

IE 692: Advanced Process Mining

Module number	IE 692
Title	Advanced Process Mining
Form of module	Lecture with exercise
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	150 h per semester
Prerequisites	Familiarity with Petri nets and BPMN for process modeling, along with basic Python programming skills, is expected. Prior experience in process mining is not a prerequisite
Aim of module	Process mining is a data-driven discipline that extracts qualitative and quantitative insights into the execution of organizational processes through the analysis of event data. It combines formal process models with data science and machine-learning techniques. The course covers the foundations, algorithms, and state-of-the-art methods of process mining, including • Process discovery • Conformance checking • Process enhancement • Predictive monitoring • Non-traditional event data, including object-centric and partially ordered event data • Event data preprocessing, abstraction, and clustering Lectures are complemented by hands-on exercises that primarily focus on pen-and-paper problem solving, supported by selected implementations and evaluations using open-source process mining tools. A group project allows students to apply and evaluate process mining and ML-based methods in a practical setting
Learning outcomes and qualification goals	Knowledge: • After completing the course, students will be familiar with both fundamental and state-of-the-art process mining techniques and algorithms for the data-driven analysis of business processes (MK1, MK2) Capabilities: • After completing the course, students will be able to analyze business processes using process mining techniques and (open-source) process mining tools (MF1, MF2, MF3) Competencies: • During the course, students will learn to apply, compare, and evaluate process mining techniques on synthetic and real-world data, both individually and in teams (MKO1, MKO3)
Media	Electronic slides and exercise sheets

Literature	• van der Aalst: <i>Process Mining: Data Science in Action</i>, 2nd edition, Springer (2016) DOI: 10.1007/978-3-662-49851-4 • van der Aalst & Carmona (Editors): <i>Process Mining Handbook</i>, Springer (2022) DOI: 10.1007/978-3-031-08848-3 • Carmona et al.: <i>Conformance Checking: Relating Processes and Models</i>, Springer (2018) DOI: 10.1007/978-3-319-99414-7 • Additional academic papers will be recommended on a per-lecture basis
Methods	• Lectures • Homework exercises to practice with joint discussion in exercise sessions
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	FSS
Person in charge	Prof. Dr. Daniel Schuster
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Social Data Science
Semester	2nd semester

IE 695: Reinforcement Learning

Module number	IE 695
Title	Reinforcement Learning
Form of module	Lecture with exercise (partially online)
Type of module	Data Analytics Methods
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 98 h per semester, including • 70 h: Lectures/exercises • 28 h: Examination preparation
Prerequisites	Machine Learning / Computer Vision / Generative Computer Vision Models course, theoretical and practical knowledge of neural networks
Aim of module	• Basic concepts of reinforcement learning: MDP, policies, on-policy, off-policy learning • Classical tabular reinforcement learning, DP, Policy Iteration, Q-Learning, SARSA, Monte-Carlo methods • Function approximation for reinforcement learning • Policy gradient methods
Learning outcomes and qualification goals	Professional competence: • After the course students will understand classical concepts of reinforcement learning as well as state-of-the-art algorithms Methodological competence: • Understand and customize popular reinforcement learning algorithms • Choose the right setting for their problem • Train agents to perform well in environments with which they interact Personal competence: • Trains abstract thinking and formal modeling of application scenarios • By solving assigned exercises independently, the transfer of the learned material to related questions is promoted
Media	Exercise sheets and lecture slides available online, blackboard
Literature	Sutton & Barto: <i>Reinforcement Learning: An Introduction</i> , 2018
Methods	Lecture, exercises every two weeks, book studies
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	HWS

Person in charge	Prof. Dr.-Ing. Margret Keuper
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	1st/ 2nd/ 3rd semester

IE 696: Advanced Methods in Text Analytics

Module number	IE 696
Title	Advanced Methods in Text Analytics
Form of module	Lecture with exercise
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 112 h per semester, including • 84 h: Pre- and post-lecture studying and revision • 28 h: Examination preparation
Prerequisites	• Fundamental notions of linear algebra and probability theory • Successful completion of “Text Analytics” (IE661), “Machine Learning” (IE675b) or “Deep Learning” (IE678)
Aim of module	This module builds upon the introduction to Natural Language Processing (NLP) from “Text Analytics” (IE661) and introduces students to cutting-edge problems, techniques, and state-of-the-art methods in NLP. The course will focus on neural models of meaning in context and present a variety of “Deep Learning” architectures for different applications in human language technology (e.g., summarization, dialogue systems, and machine translation). Moreover, we will cover open research areas, such as the explainability and interpretability of NLP models and methods to quantify the degree of bias they exhibit
Learning outcomes and qualification goals	Professional competence: • Students will acquire knowledge of state-of-the-art principles and methods of Natural Language Processing, specifically focusing on applying statistical methods to human language technologies (MK1, MK2, MF3) Methodological competence: • Successful participants will be able to understand state-of-the-art methods for Natural Language Processing and select, apply and evaluate the most appropriate techniques for various practical and application-oriented scenarios (MF3) Personal competence: -
Media	Lecture and tutorial slides, exercise sheets
Literature	Dan Jurafsky and James H. Martin: <i>Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition</i> (3rd edition, online: https://web.stanford.edu/~jurafsky/slp3/)
Methods	Lectures, tutorials
Form of assessment	Written examination

Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	FSS
Person in charge	Prof. Dr. Simone Paolo Ponzetto
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Social Data Science, Teacher Education Informatics
Semester	1st/ 2nd/ 3rd semester

CS 646: Higher Level Computer Vision

Module number	CS 646
Title	Higher Level Computer Vision
Form of module	Lecture with exercise
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 98 h per semester, including • 70 h: Lecture/exercises • 28 h: Examination preparation
Prerequisites	Basic skills in linear algebra, basic knowledge in Python and PyTorch
Aim of module	• Point Features and point matching • Object Identification • Deep Learning for Computer Vision • Object Detection • Image Segmentation • Optical Flow • Video and Motion Segmentation
Learning outcomes and qualification goals	Professional competence: • The students have a detailed understanding of Computer Vision techniques • They can evaluate given Computer Vision algorithms (MK1, MK2, MF1, MF3) Methodological competence: • Students understand the technical basis of Computer Vision algorithms • They can explain the discussed methods and implement them (MF1, MF2, MF3) Personal competence: • Understanding complex Computer Vision problems • Thorough judgment in the design and use of methods • Can work efficiently in a team (MKO1, MKO2)
Media	Exercise sheets and lecture slides are available online
Literature	• Goodfellow et al.: <i>Deep Learning</i>, MIT Press, 2016 https://www.deeplearningbook.org/ • R. Szeliski: <i>Computer Vision: Algorithms and Applications</i>, Springer, 2010. ISBN: 978-1-84882-934-3. http://szeliski.org/Book/ • D. Forsyth, J. Ponce: <i>Computer Vision: A Modern Approach</i>, Prentice Hall, 2nd ed., 2012. ISBN: 978-0136085928. • R. Hartley, A. Zisserman: <i>Multiple View Geometry in Computer Vision</i>, Cambridge University Press, 2nd ed., 2004

Methods	Lecture, weekly exercise, book studies, implementation of algorithms, visualization of results
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr.-Ing. Margret Keuper
Duration of module	1 semester
Further modules	Image Processing, Generative Computer Vision Models
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	1st/ 2nd/ 3rd semester

CS 647: Image Processing

Module number	CS 647
Title	Image Processing
Form of module	Lecture with exercise (partially online)
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 98 h per semester, including • 70 h: Lecture/exercises • 28 h: Examination preparation
Prerequisites	Basis skills in linear algebra, basis knowledge in python
Aim of module	• Introduction to Imaging (human visual system, optics, sensors) • Noise and basic operations (convolution, correlations, gradients) • Energy minimization • Variational Methods • Fourier theory and wavelet theory • Image Denoising • Segmentation • Stereo Vision
Learning outcomes and qualification goals	Professional competence: • The students have a detailed understanding of image and video processing techniques • They can evaluate given image processing algorithms (MK1, MK2, MF1, MF3) Methodological competence: • Students understand the technical basis of image processing algorithms • They can explain the discussed methods and implement them (MF1, MF2, MF3) Personal competence: • Understanding complex Image Processing problems • Thorough judgment in the design and use of methods • Can work efficiently in a team (MKO1, MKO2)
Media	Exercise sheets and lecture slides available online
Literature	• R. Szeliski: Computer Vision Algorithms and Applications, Springer, 2010. ISBN: 978-1-84882-934-3. (Online available: http://szeliski.org/Book/) • D. Forsyth, J. Ponce: Computer Vision: A Modern Approach, Prentice Hall, 2nd edition, 2012. ISBN: 978-0136085928
Methods	Lecture, weekly exercise, book studies, implementation of algorithms, visualization of results
Form of assessment	Written examination

Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	FSS
Person in charge	Prof. Dr.-Ing. Margret Keuper
Duration of module	1 semester
Further modules	Higher Level Computer Vision, Generative Computer Vision Models
Programs	B.Sc. Business Informatics, M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	1st/ 2nd/ 3rd semester

CS 668: Generative Computer Vision Models

Module number	CS 668
Title	Generative Computer Vision Models
Form of module	Lecture with exercise
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 98 h per semester, including • 70 h: Lecture/exercises • 28 h: Examination preparation
Prerequisites	Basis skills in linear algebra, basis knowledge in python/pytorch
Aim of module	• Introduction to Clustering and Unsupervised Learning • Introduction to Generative Models • Autoregressive Models for Image Generation • (Normalizing) Flow • Latent Variable Models • Latent Space Visualizations • Generative Adversarial Models • Diffusion Models • Multi-Modal Conditioning
Learning outcomes and qualification goals	Professional competence: • The students have a detailed understanding of image generation techniques, latent variable models and their evaluations (MK1, MK2, MF1, MF3) Methodological competence: • Students understand the technical basis of generative models • They can explain the discussed methods and implement them (MF1, MF2, MF3) Personal competence: • Understanding image generative models, their working principles and training procedures • Thorough judgment in the design and use of methods • Students can work efficiently in a team (MKO1, MKO2)
Media	Exercise sheets and lecture slides available online
Literature	Ian Goodfellow and Yoshua Bengio and Aaron Courville, <i>Deep Learning</i> , An MIT press book, 2016
Methods	Lecture, weekly exercise, book studies, implementation of algorithms, visualization of results
Form of assessment	Written examination

Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	FSS
Person in charge	Prof. Dr.-Ing. Margret Keuper
Duration of module	1 semester
Further modules	Higher Level Computer Vision, Genarative Computer Vision Models
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	1st/ 2nd/ 3rd semester

E. Responsible Data Science

1. Overview

Module no.	Name of Module	Offered	Language	ECTS	Page
	Legal and Ethical Aspects of Privacy	HWS	E	3	69
CS 718	AI and Data Science in Fiction and Society	HWS	E	4	71
AC 654	Additional Course - Responsible Data Science	HWS/FSS	E	max. 18	73
DS 203	Responsible AI: Conceptual Foundations, Methods and Applications	HWS	E	6	74
IS 611 (IS 6110)	Advanced Topics in Large Language Models	FSS	E	6	76

2. Detailed descriptions

Legal and Ethical Aspects of Privacy

Title	Legal and Ethical Aspects of Privacy
Form of module	Lecture
Type of module	Responsible Data Science
Level	Master
ECTS	3
Workload	Hours in presence: 28 h per semester (2 SWS) Self-study: 60 h per semester, including • 30 h: Pre- and post-lecture studying and preparation • 30 h: Examination preparation
Prerequisites	-
Aim of module	In the first section, the course acquaints the students with the notion of privacy, the challenges of conceptualizing privacy and the importance of protecting privacy. The course addresses the origins and basic principles of privacy law in Europe and contrast the European privacy foundations with the U.S. approach. In the second section, students are introduced to the European General Data Protection Regulation (GDPR) and its applicability to specific cases and basic principles. The course also addresses the question if the GDPR is suitable to protect privacy issues. In a third section the course will cover legal and ethical aspects of the use of personal and non-personal data. Data potentially allows to identify and target individuals. Sometimes this kind of individualization might be legal under the auspices of the GDPR, but the legal assessment can be more challenging when looking at it from a fundamental rights perspective. Furthermore, the question arises whether those use cases are desirable from an ethical and societal point of view. The course will use selected examples in order to illustrate the ambivalence of legality, legitimacy, and ethics. During the course, students will be familiarized with (selected) current challenges to privacy by big data, big data analytics and the use of artificial intelligence

Learning outcomes and qualification goals	Students will. . . • be aware of the challenges of conceptualising privacy and the importance of protecting privacy • have an understanding of why privacy issues are treated differently in Europe and the U.S. and the problems and weaknesses of the respective approaches • have a basic knowledge on the applicability of the General Data Protection Regulation (GDPR) and its basic principles • be aware of privacy issues and potential legal limitations when processing data • be aware of current challenges to privacy • be aware of currently discussed new approaches to privacy (e.g., privacy by design) • be aware of legal and ethical issues of using personal as well as non-personal data • have an understanding of the chances and challenges of the use of artificial intelligence for privacy
Media	Lectures, Online quizzes
Literature	Students receive reading assignments for each unit
Methods	The class will generally be conducted as a lecture
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr. Svenja Behrendt
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science, M.Sc. Social Data Science
Semester	1st/ 2nd/ 3rd semester

CS 718: AI and Data Science in Fiction and Society

Module number	CS 718
Title	AI and Data Science in Fiction and Society
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Bachelor's degree
Aim of module	In this seminar, students analyze and discuss fictional works in the area of AI and data science with respect to technological and societal aspects. The present the results orally and in a written report
Learning outcomes and qualification goals	Professional competence: - Students will learn about societal effects of AI and data science and become aware of potential threats and dangers, but also of chances of those new technologies (MK1) Methodological competence: - Students will develop methods and skills to find relevant literature for his/her topic, and to write a well-structured scientific paper and to present his/her results - He/she will be also aware of the need to avoid plagiarism - The key qualification Scientific Research is highly recommended as a prerequisite for the seminar (MF1, MF2, MF3) Personal competence: - Students will acquire skills on how to find relevant literature for a research topic, discuss a fictional work using secondary literature as background material, write a well-structured, concise paper about it and present the results of their work - He/she is well prepared to write and present a master's thesis (MKO2)
Media	Fictional and non-fictional texts
Literature	A detailed literature list is compiled for each offering
Methods	Do scientific work independently under the guidance of a professor or a research staff member
Form of assessment	- Seminar paper (15-20 pages depending on the topic) - Peer Review (up to 10 pages) - Presentation (20-30 minutes depending on the topic)
Admission requirements for assessment	-
Duration of assessment	6-8 weeks (writing of the essay) 20-30 minutes (presentation)

Language	English
Offering	HWS
Person in charge	Prof. Dr. Heiko Paulheim
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science, M.Sc. Business Informatics, Teacher Education Informatics
Semester	3rd semester

AC 654: Additional Course - Responsible Data Science

Module number	AC 654
Title	Additional Course - Responsible Data Science
Form of module	Depends on course
Type of module	-
Level	Master
ECTS	Max. 18 (A total of 18 ECTS can be recognized over all possible additional courses)
Workload	Depends on course
Prerequisites	Depends on course
Aim of module	The course falls into the responsible data science area of the MMDS and covers topics related to responsible data science but is not directly equivalent to any course in the MMDS module catalogue. The course level equals a regular course in MMDS study program. The module can be taken either at the University of Mannheim or at any other university in Germany or abroad
Learning outcomes and qualification goals	Depends on course
Media	Depends on course
Literature	Depends on course
Methods	Depends on course
Form of assessment	Depends on course
Admission requirements for assessment	Depends on course
Duration of assessment	Depends on course
Language	English preferred, but any other language possible if Mannheim faculty member is able to identify content and level
Offering	HWS/FSS
Person in charge	Lecturer at the host university
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science
Semester	2nd/ 3rd/ 4th semester

DS 203: Responsible AI: Conceptual Foundations, Methods and Applications

Module number	DS 203
Title	Responsible AI: Conceptual Foundations, Methods and Applications
Form of module	Lecture with Essay
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 28 h per semester (2 SWS) Self-study: 56 h lectures; 20 h essay / preparation oral exam
Prerequisites	Basic knowledge about AI systems (knowledge-based systems, machine learning, deep neural networks)
Aim of module	Conceptual foundations: • Understanding of important concepts in human-AI interaction and AI ethics (such as trust, autonomy, responsibility) Methods: • e.g., narrative interviews, group discussions, design research methods (prototyping, design thinking, techno-mimesis), (digital) ethnography, participatory action research Applications: • AI in medicine and healthcare • Social robotics • Generative AI • Other use cases / real-world AI applications
Learning outcomes and qualification goals	Professional competence: • Students gain insights and understanding of important concepts in human-AI interaction and AI ethics • They learn modes of transdisciplinary thinking and theorizing • Along sector-specific use cases they learn about ethical, legal and social aspects and challenges of real-world AI application, e.g. for healthcare Methodological competence: • Students learn elements of mixed-methods study design for human-AI interaction research Personal competence: • Students learn to critically assess conceptual, ethical, legal and social aspects of human-AI interaction • They gain skills in transdisciplinary research and theory-building and learn to transfer these insights to real-world human-AI interaction scenarios
Media	Slides are available online

Literature	• Voeneky, S., P. Kellmeyer, O. Mueller, and W. Burgard, ed. (2022). <i>The Cambridge Handbook of Responsible Artificial Intelligence: Interdisciplinary Perspectives</i>. Cambridge University Press. https://doi.org/10.1017/9781009207898 • Coeckelbergh, Mark. (2020). <i>AI Ethics</i>. The MIT Press. https://www.gbv.de/dms/bowker/toc/9780262538190.pdf • Heilinger, J.-C. (2022). <i>The Ethics of AI Ethics. A Constructive Critique</i>. <i>Philosophy & Technology</i>, 35(3), 61. https://doi.org/10.1007/s13347-022-00557-9 • McLennan, S., Fiske, A., Tigard, D., Müller, R., Haddadin, S., & Buyx, A. (2022). <i>Embedded ethics: A proposal for integrating ethics into the development of medical AI</i>. <i>BMC Medical Ethics</i>, 23(1), 6. https://doi.org/10.1186/s12910-022-00746-3 • Schmitt, L. (2021). <i>Mapping global AI governance: A nascent regime in a fragmented landscape</i>. <i>AI and Ethics</i>. https://doi.org/10.1007/s43681-021-00083-y
Methods	Interactive lecture
Form of assessment	Essay
Admission requirements for assessment	-
Duration of assessment	Essays need to be handed in by December 8th
Language	English
Offering	HWS
Person in charge	Jun.-Prof. Dr. Philipp Kellmeyer
Duration of module	1 semester
Further modules	Follow-up (block) seminar planned for summer semester
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	All semesters possible

IS 611 (IS 6110): Advanced Topics in Large Language Models

Module number	IS 611 (IS 6110)
Title	Advanced Topics in Large Language Models
Form of module	Lecture with exercise
Type of module	Specialization (Wifo), Responsible DS (MMDS)
Level	Master
ECTS	6
Workload	180 h per semester
Prerequisites	• Necessary: Prior knowledge of natural language processing, including familiarity with core concepts of modern language models. • Recommended: Knowledge of machine learning, AI, and computer programming
Aim of module	This course examines advanced topics in Large Language Models with a primary focus on advanced topics, for example the ethical, social, economic, organizational, and governance implications. The course explores how LLMs are deployed, evaluated, governed, and contested in real-world settings. Particular emphasis is placed on questions such as bias and fairness, transparency and accountability, privacy and surveillance, misinformation and manipulation, intellectual property, safety and misuse, inequalities in access and power, the transformation of knowledge work, and the role of LLMs in societies and organisations. The course may also address adjacent topics such as regulation, platform power, sustainability, human-AI interaction, cultural effects, and methodological challenges in studying LLMs in society. The exercises part of the course deepens the material through case discussions, critical reading of current research, applied analyses, and project-based work
Learning outcomes and qualification goals	Upon successful completion of this course, students: • Can critically assess the ethical challenges arising from the development and deployment of Large Language Models • Can analyze the social consequences of LLMs for communication, education, work, public discourse, and institutions • Can assess risks related to bias, fairness, privacy, transparency, safety, misuse, and accountability • Can apply conceptual and critical tools to examine real-world use cases of LLM-based systems • Can develop well-founded positions on the responsible design, deployment, and governance of LLMs
Media	-
Literature	Current research literature and case materials will be provided during the course
Methods	-

Form of assessment	Written examination
Admission requirements for assessment	Students have to submit exercises and short assignments and collect at least 50% of the available points to be admitted to the exam
Duration of assessment	60 minutes
Language	English
Offering	FSS
Person in charge	Prof. Dr. Markus Strohmaier, N.N.
Duration of module	1 semester
Further modules	-
Programs	Mannheim Master in Management, M.Sc. Economic and Business Education, M.Sc. Economics, M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Social Data Science
Semester	-

F. Data Science Applications

1. Overview

Module no.	Name of Module	Offered	Language	ECTS	Page
AC 655	Additional Course - Data Science Applications	HWS/FSS	E	max. 18	79
IE 655	Industrial Applications of Artificial Intelligence	FSS	E	9	81
IE 698	Foundations and Applications of Digital Health Technologies	FSS	E	3	83
IE 699	Co-creating digital health applications with design methodology	FSS	E	6	85
CS 630	Generative Software Engineering	FSS	E	6	87
DS 601	AI product developmen	FSS	E	6	89
IS 540 (IS 5400)	Management of Enterprise Systems	HWS	E	6	MMM*
MKT 545 (MKT 5450)	Customers, Markets and Firm Strategy	FSS	E	6	MMM*
MKT 511 (MKT 5110)	Marketing Analytics	FSS	E	6	MMM*
	Elective Master of Science Psychologie, School of Social Sciences	HWS/FSS	E	4	M.Sc. Psy*

* For a detailed description, please see the module catalogue of the respective following degree programs:

- MBE: M.Sc. Mathematics in Business and Economics (only available in German)
<https://www.wim.uni-mannheim.de/studium/studienorganisation/m-sc-wirtschaftsmathematik/>
- MMM: Mannheim Master in Management
<https://www.bwl.uni-mannheim.de/en/module-catalog-mmm/>
- M.Sc. Psy: Master of Science Psychologie - Arbeit, Wirtschaft, Gesellschaft (only in German)
<https://www.sowi.uni-mannheim.de/en/academics/students/psychology/msc-in-psychology-work-economy-and-society/>

2. Detailed descriptions

AC 655: Additional Course - Data Science Applications

Module number	AC 655
Title	Additional Course - Data Science Applications
Form of module	Typically: lecture, other forms of modules might be credited upon request
Type of module	-
Level	Master
ECTS	Max. 18 (The individual modules can vary between 2 and 10 ECTS. A total of 18 ECTS can be recognized as additional courses)
Workload	The workload and its distribution on different activities are determined by the credited course. For courses taken at the University of Mannheim, they can be found in the respective module catalogue. One ECTS credit equals 30 hours of work
Prerequisites	The prerequisites are determined by the credited course. For courses taken at the University of Mannheim, they can be found in the respective module catalogue
Aim of module	The course falls into the data science applications area of the MMDS and covers topics related to data science applications, but is not directly equivalent to any course in the MMDS module catalogue. The course level equals a regular course in MMDS study program. The module can be taken either at the University of Mannheim or at any other university in Germany or abroad
Learning outcomes and qualification goals	The learning outcomes and qualification goals are determined by the credited course. For courses taken at the University of Mannheim, they can be found in the respective module catalogue
Media	Determined by the credited course. For courses taken at the University of Mannheim, they can be found in the respective module catalogue
Literature	Determined by the credited course. For courses taken at the University of Mannheim, they can be found in the respective module catalogue
Methods	Determined by the credited course. For courses taken at the University of Mannheim, they can be found in the respective module catalogue
Form of assessment	Typically a written or oral examination, but can also include other forms of examination. Details are determined by the examination modalities of the credited course
Admission requirements for assessment	-
Duration of assessment	-

Language	English preferred, but any other language possible if Mannheim faculty member is able to identify content and level
Offering	HWS/FSS
Person in charge	Lecturer at the host university
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science
Semester	2nd/ 3rd/ 4th semester

IE 655: Industrial Applications of Artificial Intelligence

Module number	IE 655
Title	Industrial Applications of Artificial Intelligence
Form of module	Lecture and accompanying tutorials
Type of module	Specialization Course
Level	Master
ECTS	9
Workload	Hours in presence: 56 h per semester (2+2 SWS) Self-study: 214 h per semester, including the creation of a learning portfolio
Prerequisites	Necessary Knowledge: • Machine Learning Concepts and Techniques • Programming in Python
Aim of module	Participants will learn about the use of Artificial Intelligence methods, mostly from the field of machine learning in different sectors and industries. They will learn about application areas in the primary, secondary and tertiary sector, get an introduction to examples of such applications that have been published on a scientific level and gather some experience in working with data from the respective fields using publically available datasets
Learning outcomes and qualification goals	Professional competence: • Students will acquire knowledge about possible applications of machine learning in different branches of industry as well as the dominant methods used in these areas (MK2, MK3) Methodological competence: • Identify potential for applying AI methods in different areas of industry • Decide on a suitable method for addressing typical problems in these industries (MF2) Personal competence: • Participants will learn to reflect and document their own learning process (MKO2)
Media	Slides, Data Sets, Software Tools
Literature	Various Scientific Publications - details in the lecture slides
Methods	Lectures, tutorials, independent study
Form of assessment	Learning Portfolio
Admission requirements for assessment	-
Duration of assessment	Continuous assessment throughout the module (learning portfolio)
Language	English

Offering	FSS
Person in charge	Prof. Dr. Heiner Stuckenschmidt
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Social Data Science
Semester	2nd/ 3rd/ 4th semester

IE 698: Foundations and Applications of Digital Health Technologies

Module number	IE 698
Title	Foundations and Applications of Digital Health Technologies
Form of module	Lecture
Type of module	Specialization Course
Level	Master
ECTS	3
Workload	Hours in presence: 28 h per semester Self-study: 56 h per semester
Prerequisites	Necessary: • Basic knowledge about AI systems (knowledge-based systems, machine learning, deep neural networks) Optional: • Participation of Lecture “Responsible AI” Winter Term 2023/24 • Basic knowledge about digital health systems
Aim of module	Conceptual foundations: • Understanding of important concepts in digital health (eHealth, mHealth, telehealth, digital phenotyping, digital twins, and other concepts) Methods: • e.g., interviewing, group discussions, design research methods (prototyping, design thinking, co-design), (digital) ethnography Applications: • AI in digital health apps in medicine and healthcare • AI-based robots • AI-based VR systems
Learning outcomes and qualification goals	Knowledge: • Students gain insights and understanding of important concepts in digital health • They learn modes of transdisciplinary thinking and theorizing on digital health (MK1) Capabilities: • Students learn elements of mixed-methods study design for digital health research and co-design methodology • They learn to assess scientific publications in this domain critically (MF1, MF2) Competencies: • Students learn to critically assess the conceptual, technical, ethical, legal and social aspects of digital health applications (MKO1)
Media	Slides and supporting media will be available online

Literature	• Fagherazzi, G. (2020). <i>Deep Digital Phenotyping and Digital Twins for Precision Health: Time to Dig Deeper</i>. Journal of Medical Internet Research, 22, e16770. https://www.jmir.org/2020/3/e16770/ • Hahn, H. & Schreiber, A. (2019). <i>E-Health</i>. In Neugebauer, R. (Ed.), <i>Digital Transformation</i> (pp. 311-334). Springer. https://doi.org/10.1007/978-3-662-58134-6_19 • Budd, J. et al. (2020). <i>Digital technologies in the public-health response to COVID-19</i>. Nat Med, 26, 1183-1192. https://www.nature.com/articles/s41591-020-1011-4 • Voeneky, S., Kellmeyer, P., Mueller, O., & Burgard, W. (Eds.). (2022). <i>The Cambridge Handbook of Responsible Artificial Intelligence: Interdisciplinary Perspectives</i>. Cambridge University Press. https://doi.org/10.1017/9781009207898
Methods	Interactive lecture
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	45 minutes
Language	English
Offering	FSS
Person in charge	Jun.-Prof. Dr. Philipp Kellmeyer
Duration of module	1 semester
Further modules	Lecture on Responsible AI, Fall 2024
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	All semesters possible

IE 699: Co-creating Digital Health Applications with Design Methodology

Module number	IE 699
Title	Co-creating digital health applications with design methodology
Form of module	Exercise
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester Self-study: 112 h per semester
Prerequisites	Necessary: • Basic knowledge about AI systems (knowledge-based systems, machine learning, deep neural networks) Optional: • Participation of Lecture “Responsible AI” Winter Term 2023/24 • Basic knowledge about digital health systems
Aim of module	Conceptual foundations: • Understanding of important concepts in digital health (eHealth, mHealth, telehealth, digital phenotyping, digital twins and other concepts) Methods: • Interviewing, group discussions, design research methods (prototyping, design thinking, co-design), (digital) ethnography Applications: • By-design approaches to developing digital health apps and other solutions
Learning outcomes and qualification goals	Knowledge: • Along with digital health use cases, they learn basic by-design principles for developing digital health applications (MK1) Capabilities: • Students learn elements of mixed-methods study design for digital health research and co-design methodology via design fictions and mockups developed in groups in the exercise (MF1, MF2, MF3) Competencies: • Students learn to apply their knowledge and capabilities regarding by-design approaches to solve specific problems in a team context (MKO1)
Media	Slides and supporting media will be available online

Literature	• Fagherazzi, G. (2020). <i>Deep Digital Phenotyping and Digital Twins for Precision Health: Time to Dig Deeper</i>. Journal of Medical Internet Research, 22, e16770. https://www.jmir.org/2020/3/e16770/ • Hahn, H. & Schreiber, A. (2019). <i>E-Health</i>. In Neugebauer, R. (Ed.), <i>Digital Transformation</i> (pp. 311-334). Springer. https://doi.org/10.1007/978-3-662-58134-6_19 • Budd, J. et al. (2020). <i>Digital technologies in the public-health response to COVID-19</i>. Nat Med, 26, 1183-1192. https://www.nature.com/articles/s41591-020-1011-4 • Voeneky, S., Kellmeyer, P., Mueller, O., & Burgard, W. (Eds.). (2022). <i>The Cambridge Handbook of Responsible Artificial Intelligence: Interdisciplinary Perspectives</i>. Cambridge University Press. https://doi.org/10.1017/9781009207898
Methods	Exercise
Form of assessment	Presentation in course
Admission requirements for assessment	-
Duration of assessment	20 minutes (15 minutes Presentation, 5 minutes Q&A)
Language	English
Offering	FSS
Person in charge	Jun.-Prof. Dr. Philipp Kellmeyer
Duration of module	1 semester
Further modules	Lecture on Responsible AI, Fall 2024
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	All semesters possible

CS 630: Generative Software Engineering

Module number	CS 630
Title	Generative Software Engineering
Form of module	Lecture, exercises, tutorials and projects
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (2+2 SWS) Self-study: 112 h per semester, including • 28 h: Preparation and studying of lectures/tutorials • 84 h: Preparation of the exams
Prerequisites	• Software Testing and Experimentation • Knowledge of Java and/or Python
Aim of module	The course introduces the fundamental principles, practices and applications of generative software engineering (GSE) as well as the supporting models, tools and services: • Software Recommendation and Reuse • Code Search Engines • Principles of GSE • Existing GSE models, tools and services • Defining GSE “problems” • Applications of GSE • Evaluation and comparison of GSE models, tools and services
Learning outcomes and qualification goals	Professional competence: • Students will be familiar with the principles, practices and applications of GSE as well as supporting models, tools and services (MK1, MK2) Methodological competence: • Students will have the ability to judge, select and apply GSE techniques and tools in practical software engineering • They will be able to understand academic GSE publications and perform research in GSE (MF1, MF2, MF3) Personal competence: • Students will have improved skills in analytical thinking and applying theoretical knowledge to solve practical problems, especially in the use of AI to enhance software engineering (MKO1, MKO3)
Media	Students acquire knowledge of the principles, practices and applications of GSE, including supporting models, tools and services and methods, through lectures and tutorials. The students apply their knowledge individually and in team projects and report their findings in written reports and oral presentations
Literature	-
Methods	Lectures, weekly exercises, experimentation with different systems

Form of assessment	• Project reports (70%) • Oral presentations (30%)
Admission requirements for assessment	-
Duration of assessment	-
Language	English
Offering	FSS
Person in charge	Dr. Marcus Kessel
Duration of module	1 semester
Further modules	Model-Driven Development, Software Testing and Experimentation
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	2nd/ 3rd/ 4th semester

DS 601: AI product development

Module number	DS 601
Title	AI product development
Form of module	Lecture and Tutorial
Type of module	Foundations of AI product development
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 124 h per semester, including • 40 h: Practicing class exercises • 84 h: Working on class group project
Prerequisites	To excel in this course, students should have a solid foundation in data science, data analysis, and machine learning algorithms. A basic understanding of Large Language Models (LLMs) is essential. Working knowledge of advanced AI concepts like Retrieval-Augmented Generation (RAG), various prompting methods, and practical experience in developing prototypes with AI tools will be highly beneficial. Prior to admission, students must submit a one-page, single spaced statement of purpose. This will share their interest in this project, summarize relevant coursework and grades, experience, and ideas. Students can apply on one of two tracks: as software engineers or business entrepreneurs. Please state clearly which track is being selected and what academic program and degree you are currently pursuing. Groups will integrate members from each track, integrating product development with software design and implementation. A willingness to focus in one area but work and learn across both is essential. Details for the application submission will be on the website of Prof. Ratkovic's Chair
Aim of module	The primary aim of this course is to equip students with the skills to transform their project ideas into viable business products, all while maintaining a steadfast focus on the core principles of responsible computing and explainable AI. Students will collaborate with organizations in technology, policy, and government, developing projects that advance key areas like journalism, fact-checking, digital literacy, and healthcare. The program provides continuous mentoring and access to vital resources and networks, enabling participants to move beyond traditional academic boundaries and engage in real-world product development

Learning outcomes and qualification goals	• Technology-oriented students learn the concepts, algorithms and strategies used to solve concrete, practical application-oriented problems in informatics (MK2) • Independently tackle problems in data management and analytics and describe their results in a structured, written form (MF4) • Apply their knowledge and capabilities to solve specific problems in a team context (MKO1) • Use their interdisciplinary education to mediate between technical and non-technical individuals (MKO2) • Evaluate the latest changes in programming languages, systems, models and, wherever possible, exploit them to develop better solutions to data-science related problems (MKO3)
Media	Lecture slides and exercises will be available online
Literature	Since the course emphasizes practical applications and AI product development, project-specific literature will be provided during the sessions alongside the presentation slides
Methods	Lecture elements, weekly product development updates
Form of assessment	• Final product (report or dashboard; 70%) • Preliminary oral or written reports (20%) • Comments and feedback on other groups (10%)
Admission requirements for assessment	Homework, practical programming projects, or written and oral reports, in-class exam
Duration of assessment	-
Language	English
Offering	FSS
Person in charge	Prof. Dr. Marc Ratkovic
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Social Data Science, M.Sc. Data Science, Mannheim Master in Management, CDSS
Semester	2nd/ 4th semester

G. Projects and Seminars

1. Overview

Module no.	Name of Module	Offered	Language	ECTS	Page
AC 653	Additional Course - Projects and Seminars	HWS/FSS	E	max. 18	92
TP 500	Team Project	HWS/FSS	G/E	12	93
IP 500	Individual Project	HWS/FSS	G/E	8	95
SQ 500	Scientific Research	HWS/FSS	E	2	97
CS 701	Seminar Selected Topics in Algorithmics and Cryptography	Irregular	E	4	99
CS 704	Master Seminar Artificial Intelligence	Irregular	E	4	101
CS 707	Data Analytics Seminar	Irregular	E	4	103
CS 708	Seminar Software Engineering	Irregular	E	4	105
CS 709	Seminar Text Analytics	Irregular	G/E	4	107
CS 710	Seminar Selected Topics in Data Science	Irregular	G/E	4	109
CS 715	Solving Complex Tasks using Large Language Models	Irregular	E	4	111
CS 716	IT-Security	Irregular	E	4	113
CS 717	Master Seminar on Computer Vision	HWS/FSS	E	4	115
CS 720	Uncertainty Estimation	Irregular	E	4	117
CS 721	Seminar Data-Science I	Irregular	E	4	119
CS 733	Advanced Topics in Process Mining	FSS	E	4	121
CS 735	Computational Analysis of Communication	HWS/FSS	E	4	123
IE 704	Seminar AI Systems Engineering	Irregular	E	4	125
IS 723 (IS 7230)	Seminar Data-Science II (Empirical Studies)	HWS/FSS	E	4	MMM*

* For a detailed description, please see the module catalogues of the respective following degree programs:

- MMM: Mannheim Master in Management
<https://www.bwl.uni-mannheim.de/en/module-catalog-mmm/>

2. Detailed descriptions

AC 653: Additional Course - Projects and Seminars

Module number	AC 653
Title	Additional Course - Projects and Seminars
Form of module	Depends on course
Type of module	Depends on course
Level	Master
ECTS	Max. 18(A total of 18 ECTS can be recognized over all possible additional courses)
Workload	Depends on course
Prerequisites	Depends on course
Aim of module	The course equals a seminar in the MMDS study program. The module can be taken either at the University of Mannheim or at any other university in Germany or abroad
Learning outcomes and qualification goals	Depends on course
Media	Depends on course
Literature	Depends on course
Methods	Depends on course
Form of assessment	Depends on course
Admission requirements for assessment	Depends on course
Duration of assessment	Depends on course
Language	English preferred, but any other language possible if Mannheim faculty member is able to identify content and level
Offering	HWS/FSS
Person in charge	Lecturer at the host university
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science
Semester	2nd/ 3rd/ 4th semester

TP 500: Team Project

Module number	TP 500
Title	Team Project
Form of module	Project
Type of module	Team Project
Level	Master
ECTS	12 in two consecutive semesters (12 month) or 12 in one semester (6 month)
Workload	Hours in presence: • 12 month-project: 28 h per semester (2 SWS) • 6 month-project: 56 h per semester (4 SWS) Self-study (12 month project): 140 h per semester • 112 h: Pre- and post-lecture studying, revision and free self-study • 28 h: Preparation of examination/presentation Self-study (6 month project): 280 h per semester • 224 h: Pre- and post-lecture studying, revision and free self-study • 56 h: Preparation of examination/presentation
Prerequisites	Depends on topic
Aim of module	The students solve a practical problem as a team. The participants have to analyze and refine the problem and come up with a project plan for developing a concrete solution that will be carried out by the team over the duration of a whole year. Concrete topics for projects are defined by the supervisors and offered to the students who can apply for different topics. Problem area and techniques involved depend on the expertise of the offering chair
Learning outcomes and qualification goals	Professional competence: Depending on the actual topic of the project, participants will acquire: • In-depth knowledge in a certain application of business informatics and data science • Knowledge about methods and technologies typically applied in the application area • Knowledge about practical problems and challenges when applying a certain technique in a given application area (MK1, MK2, MK3) Methodological competence: Participants will learn to: • Refine a given problem statement by analyzing requirements and the state of the art using techniques like literature research and expert interviews • Define a workplan including tasks, milestones, deliverables and resources and continually assess and modify the plan according to the actual progress of the work (MF1, MF2, MF3, MF4)

	Personal competence: Being a team effort, the project explicitly targets personal competence in terms of: • Working in and managing a team of experts possibly from different academic and cultural backgrounds • Taking part in discussions and learning to contribute the own opinion without overruling other opinions • Self-management and responsibility within the requirements of collaborative work (MKO1, MKO2, MKO3)
Media	Depends on project
Literature	Depends on topic
Methods	Team-discussions, Presentations, Teamwork, Individual preparation of empirical contributions, self-study
Form of assessment	• One or more reports totaling 50-120 pages (depending on the topic) • Presentation (10-90 minutes)
Admission requirements for assessment	• 12-month project: withdrawal within the first 6 weeks possible without failing • 6-month project: withdrawal within the first 3 weeks possible without failing
Duration of assessment	• Final report (length: one or more reports totaling 50-120 pages depending on the topic) • Presentation (10-90 minutes depending on the topic)
Language	English
Offering	HWS/FSS
Person in charge	Professors of the Institute of Computer Science and Business Informatics or of the Area Information Systems of the Business School
Duration of module	1 semester or 2 semesters
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	1st/ 2nd/ 3rd semester

IP 500: Individual Project

Module number	IP 500
Title	Individual Project
Form of module	Project
Type of module	Individual Project
Level	Master
ECTS	8
Workload	Self-study: 240 h per semester
Prerequisites	Depends on topic
Aim of module	The student solves a practical problem individually. The student has to analyse and refine the problem and come up with a project plan for developing a concrete solution. Concrete topics for projects are defined by the supervisors and offered to the students who can apply for different topics. Problem area and techniques involved depend on the expertise of the offering chair
Learning outcomes and qualification goals	Professional competence: Depending on the actual topic of the project, participants will acquire: • In-depth knowledge in a certain application of data science • Knowledge about methods and technologies typically applied in the application area • Knowledge about practical problems and challenges when applying a certain technique in a given application area (MK1, MK2, MK3) Methodological competence: Participants will learn to: • Refine a given problem statement by analysing requirements and the state of the art using techniques like literature research and expert interviews • Define a workplan including tasks, milestones, deliverables and resources and continually assess and modify the plan according to the actual progress of the work (MF1, MF2, MF3, MF4, MKO3) Personal competence: -
Media	Depends on project
Literature	Depends on topic
Methods	Self-study, presentations
Form of assessment	• Final report (length: 10-50 pages depending on the topic) • Presentation (15-30 minutes depending on the topic)
Admission requirements for assessment	-
Duration of assessment	• 6 months (execution of the project and writing of the report) • 15-30 minutes (presentation)

Language	English/German
Offering	Irregular
Person in charge	Professors of the Institute of School of Business Informatics and Mathematics of the School of Social Sciences
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science
Semester	1st/ 2nd/ 3rd semester

SQ 500: Scientific Research

Module number	SQ 500
Title	Scientific Research
Form of module	Inverted Classroom with Exercises
Type of module	Key Qualification
Level	Master
ECTS	2
Workload	80% (48 hours) self-study and 20% (12 hours) coursework
Prerequisites	-
Aim of module	This course focuses on the basic key competences that are needed to successfully write a scientific paper or a thesis. It is recommended that you take this module before you participate in a seminar. Topics include: • Scientific process and scientific writing • Bibliographic research methodology • Search strategies in bibliographic databases • Finding data for your research • How to read, understand and cite scientific literature • Reference management systems and L^AT_EX
Learning outcomes and qualification goals	Professional competence: • The students understand how to work scientifically and how to write a thesis Methodological competence: • The students can find relevant publications for a research question Personal competence: • Everybody wrote a short overview of his/her research question • Everybody installed and used exemplary tools to support the work process
Media	-
Literature	• <i>The Craft of Research</i> / Wayne C. Booth; Gregory G. Colomb; Joseph M. Williams (Chicago Guides to Writing, Editing, and Publishing); 3rd ed.; University of Chicago Press, 2008; XVII, 317 pages; illustrations; 22cm • L^AT_EX (<i>Wikibook</i>): http://en.wikibooks.org/wiki/LaTeX
Methods	The students can individually work through the video recordings of the lecture and do the accompanying hands-on exercises. The live sessions will be held online as well and are mainly dedicated to clarifying any questions from the students
Form of assessment	Eine digital unterstützte Hausarbeit
Admission requirements for assessment	Successful participation in at least 60% of all online exercises

Duration of assessment	150 minutes
Language	English
Offering	HWS/FSS
Person in charge	Lecturer from the University Library (UB)
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	All semesters possible

CS 701: Seminar Selected Topics in Algorithmics and Cryptography

Module number	CS 701
Title	Selected Topics in Algorithmics and Cryptography
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Algorithmics (CS 550) or Cryptography II (CS 651) or Courses in Algorithms or Cryptography or Theoretical Computer Science or Complexity Security at BSc or MSc Level, key qualification scientific research
Aim of module	The students prepare a scientific report on a current research topic on the basis of published papers under guidance of a scientific staff member and give a presentation. The topic will be proposed by the professor, but the students may also propose topics. Active participation in the seminar presentations of fellow students will be expected
Learning outcomes and qualification goals	Professional competence: • The students gain a deep understanding of the research topic, are able to explain the topic in detail in a clean and transparent way and are able to classify the significance of the topic and the results in relation to the current state of research in the corresponding research area Methodological competence: • The students are able to read, understand and explore scientific literature relevant to the topic • They are aware of the need to avoid plagiarism Personal competence: • The student has learned how to find relevant literature for a research topic, write a well-structured and clear report about it and give a presentation • The seminar serves also as preparation for writing and presenting the master's thesis
Media	Scientific papers and books, Presentation systems like PowerPoint or projector L ^A T _E X
Literature	Depends on the topic
Methods	Do scientific work independently under the guidance of a research staff member and manage an active discussion on the topic in a group of peers
Form of assessment	• Presentation and discussion (15-60 minutes) • Written essay/report (5-25 pages) • Participation

Admission requirements for assessment	Timely hand-in of seminar papers and presentation materials
Duration of assessment	• 60 minutes presentation with 15-30 minutes discussion • Written report (length: 15 pages)
Language	English
Offering	FSS
Person in charge	Prof. Dr. Matthias Krause
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Mathematics in Business and Economics
Semester	3rd semester

CS 704: Master Seminar Artificial Intelligence

Module number	CS 704
Title	Master Seminar Artificial Intelligence
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Decision Support or Data Mining or Knowledge Management
Aim of module	The student prepares a scientific paper and gives a presentation on a current research topic based on published research. State-of-the-art topics are proposed by the professors. The paper and the presentation are prepared under the guidance of a professor or a research staff member. Active participation in the seminar discussions is expected
Learning outcomes and qualification goals	Professional competence: • The student gains a deep understanding of the research topic • He/she is able to describe/summarize the topic in detail in his/her own words • He/she reflects on the topic and judges the contribution of the research papers (MK1, MK2, MK3) Methodological competence: • The student is able to write a well-structured scientific paper and to present his/her results • He/she is also aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar (MF2) Personal competence: • The student has learned how to write a well-structured, concise paper and give a presentation • This is part of the preparation to write and present a master's thesis
Media	Scientific papers and books, final presentation with L ^A T _E X slides
Literature	Depends on the topic of the seminar
Methods	Do scientific reading independently under the guidance of a professor or a research staff member, active discussions in a group of peers
Form of assessment	We offer two types of seminars: • <i>Seminar with a focus on theory and literature:</i> Two presentations (30 minutes overall) and a final seminar report (15-20 pages) • <i>Practical Seminar:</i> Two presentations (30 minutes overall) and a final seminar report (12-15 pages) presenting own results

	Offered in exceptional cases as group work. In such cases, the practical task is more extensive/challenging. Code has to be submitted in addition to the final report. More about content and type of the seminar can be found at https://www.uni-mannheim.de/dws/teaching/course-details/courses-for-master-candidates/
Admission requirements for assessment	-
Duration of assessment	• Both presentations must be given within a span of 5-10 weeks • Final report has to be submitted after this period
Language	English
Offering	Irregular
Person in charge	Prof. Dr. Heiner Stuckenschmidt
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	3rd semester

CS 707: Data Analytics Seminar

Module number	CS 707
Title	Data Analytics Seminar
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	• CS Fundamentals • Relevant Master Courses (depending on the topic) • Key qualification Scientific Research
Aim of module	Explore, present and critically discuss recent research literature in the areas of machine learning and/or data processing. The precise topics vary from term to term
Learning outcomes and qualification goals	Professional competence: • The student gains a deep understanding of the research topic Methodological competence: • The student is able to read, understand, explore, present, defend and critically discuss scientific literature relevant to his/her topic • He/she is also aware of the need to avoid plagiarism Personal competence: • The seminar fosters critical thinking and discussion skills • Students will be prepared to research and critically assess current scientific literature in preparation of their Master's thesis
Media	Scientific papers and books, software and datasets, presentations
Literature	Depends on the topic of the seminar
Methods	Do scientific work independently under the guidance of a professor or a research staff member, active discussions in a group of peers
Form of assessment	Individual grading of the oral presentations, experimental results (if applicable), active participation in the seminar's discussion rounds, a written discussion summary, and timeliness of hand-ins
Admission requirements for assessment	-
Duration of assessment	N/A
Language	English
Offering	HWS/FSS (typically)
Person in charge	Prof. Dr. Rainer Gemulla
Duration of module	1 semester

Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	3rd semester

CS 708: Seminar Software Engineering

Module number	CS 708
Title	Seminar Software Engineering
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Bachelor's degree, the fundamentals
Aim of module	Student prepares a scientific paper and gives a presentation on a current software engineering research topic based on published research papers. State-of-the-art topics are proposed by the software engineering group. Active participation in the seminar discussions is expected
Learning outcomes and qualification goals	Professional competence: • The student gains a deep understanding of the research topic • They are able to describe/summarize the topic in detail in their own words • They reflect on the topic and judge the contribution of the research papers Methodological competence: • The student is able to find the relevant literature for their topic, to write a well-structured scientific paper and to present their results • They are also aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar Personal competence: • The student has learned how to find relevant literature for a research topic, write a well-structured, concise paper about it and give a presentation
Media	Scientific papers and books, presentation
Literature	Depends on the topic of the seminar
Methods	Scientific work performed independently under the guidance of a member of the software engineering group
Form of assessment	• Seminar paper (final report): 12-13 pages excluding references • Presentation (15 minutes) plus discussion (5-10 minutes) • Peer review: 1-5 pages • Accompanying work products (e.g., code, models, spreadsheets, documents etc.)
Admission requirements for assessment	-

Duration of assessment	Presentation (15 minutes) plus discussion (5-10 minutes)
Language	English
Offering	Irregular
Person in charge	Prof. Dr. Colin Atkinson
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	1st/ 2nd/ 3rd semester

CS 709: Seminar Text Analytics

Module number	CS 709
Title	Seminar Text Analytics
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	IS 661 (IS 6610) “Text Analytics” or IE 663 “Web Search and Information Retrieval” or IE 696 “Advanced Text Analytics”, Fundamental notions of linear algebra and probability theory
Aim of module	In this seminar, students write a survey/scientific paper and provide an overview presentation of state-of-the-art research, as found within the existing literature (i.e., published research papers). Topics of interest focus around a variety of problems and tasks from the fields of Natural Language Processing and Information Retrieval. The paper and the presentation are prepared under the guidance of a professor or a research staff member
Learning outcomes and qualification goals	Professional competence: • Students will acquire a deep understanding of the research topic • They are expected to describe in-depth and summarize the topic in detail in their own words, as well as to judge the contribution of the research papers to ongoing research Methodological competence: • Students will develop methods and skills to find relevant literature for their topic, to write a well-structured survey/scientific paper and to present their results • They will also be aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar Personal competence: • Students will acquire skills on how to find relevant literature for a research topic, write a well-structured, concise paper about it and present the results of their work • They are well prepared to write and present a master’s thesis
Media	Scientific papers and books, presentation with PowerPoint or L ^A T _E X
Literature	Depends on the topic of the seminar
Methods	Do scientific work independently under the guidance of a professor or a research staff member, active discussions in a group of peers

Form of assessment	• Flash presentation (3 minutes) • Final presentation (15 minutes presentation + discussion + moderation) • Report (10 pages + references/appendix) • Peer review • Optionally: code hand-in
Admission requirements for assessment	-
Duration of assessment	-
Language	English or German
Offering	Irregular
Person in charge	Prof. Dr. Simone Paolo Ponzetto
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	3rd semester

CS 710: Seminar Selected Topics in Data Science

Module number	CS 710
Title	Selected Topics in Data Science
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Bachelor's degree, the fundamentals
Aim of module	In this seminar, students perform scientific research, either in the form of a literature review or by conducting a small experiment, or a mixture of both, and prepare a written report about the results. Topics of interest focus around a variety of problems and tasks from the fields of Data Mining, Web Mining, or the Semantic Web
Learning outcomes and qualification goals	Professional competence: • Students will acquire a deep understanding of the research topic • He/she is expected to describe in-depth and summarize the topic in detail in his/her own words, as well as to judge the contribution of the research papers to ongoing research (MK1, MK2, MK3) Methodological competence: • Students will develop methods and skills to find relevant literature for his/her topic, to prepare methodologically sound scientific experiments, and to write a well-structured scientific paper and to present his/her results • He/she will also be aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar (MF2) Personal competence: • Students will acquire skills on how to find relevant literature for a research topic, organize a small research task, write a wellstructured, concise paper about it and present the results of their work • He/she is well prepared to write and present a master's thesis (MF2)
Media	Scientific papers and books
Literature	Depends on the topic of the seminar
Methods	Do scientific work independently under the guidance of a professor or a research staff member
Form of assessment	• Seminar paper (5-25 pages depending on the topic) • Peer Review (1-10 pages depending on the topic) • Presentation (15-60 minutes depending on the topic)
Admission requirements for assessment	-

Duration of assessment	• 6-8 weeks (writing of the essay) • 15-60 minutes (presentation)
Language	English or German
Offering	Irregular
Person in charge	Prof. Dr. Heiko Paulheim
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	3rd semester

CS 715: Solving Complex Tasks using Large Language Models

Module number	CS 715
Title	Solving Complex Tasks using Large Language Models
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Bachelor's degree, the fundamentals
Aim of module	The seminar explores prompt engineering, fine-tuning, as well as agent-based techniques for enabling LLMs to handle complex tasks as well as using LLMs to evaluate complex outputs. The seminar features literature as well as experimental topics. The goal of the literature topics is to summarize the state of the art concerning the application and evaluation of LLMs for a specific task. The goal of the experimental topics is to verify the utility of advanced techniques by applying them to tasks beyond the tasks used in the original papers
Learning outcomes and qualification goals	Professional competence: • Students will acquire a deep understanding of the state of the art with respect to their research topic • They learn to summarize the state of the art, judge methods along different criteria, design experiments, and present experimental results Methodological competence: • Students will develop methods and skills to find relevant literature for their topic, to prepare methodologically sound scientific experiments and to write a well-structured scientific paper and to present their results • They will also be aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar Personal competence: • Students will acquire skills on how to find relevant literature for a research topic, organize a small research task, write a wellstructured, concise paper about it and present the results of their work
Media	Scientific papers and books
Literature	Depends on the specific topic
Methods	Perform scientific work independently under the guidance of a professor or a research staff member
Form of assessment	• Report (12-15 pages excluding references) • Presentation (12 minutes + 8 minutes discussion)

Admission requirements for assessment	-
Duration of assessment	6-8 weeks (writing of the seminar paper)
Language	English
Offering	Irregular
Person in charge	Prof. Dr. Christian Bizer
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Social Data Science, Teacher Education Informatics
Semester	3rd semester

CS 716: IT-Security

Module number	CS 716
Title	IT-Security
Form of module	Block seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Bachelor's degree, the fundamentals
Aim of module	Seminar participants choose a topic from a given list of topics or, if applicable, based on their own suggestions. They prepare a presentation on this topic and write a corresponding paper. They also conduct literature research and systemize the current state of research. The focus here is on working out and understanding the scientific content and being able to classify and evaluate it. All participants give a presentation. Active participation in the seminar is expected
Learning outcomes and qualification goals	Professional competence: • Students have in-depth knowledge in the field covered by the seminar Methodological competence: • Students are able to find relevant, current literature on a research topic, thoroughly understand a research topic, process it both in terms of content and didactics, and present it to others. They will be well prepared to write and present a Master's Thesis Personal competence: • Students are able to work independently • They have learned how to find relevant literature on a research topic, delve deeper into a subject, write a well-structured, concise report on it, and give a presentation
Media	Scientific papers and books, final presentation with PowerPoint or similar software
Literature	Depends on the topic of the seminar
Methods	Independent academic work and preparation of a presentation (lecture and paper) in collaboration with a professor or research assistant. Active discussions in a group of fellow students
Form of assessment	• Presentation • Technical discussion • Seminar paper (8 pages, double-column)
Admission requirements for assessment	Timely hand-in of seminar paper and presentation
Duration of assessment	• 90 minutes for presentation and technical discussion • Seminar paper (8 pages, double-column)

Language	English
Offering	Irregular
Person in charge	Prof. Dr. Frederik Armknecht
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, Teacher Education Informatics, M.Sc. Mathematics in Business and Economics
Semester	3rd semester

CS 717: Master Seminar on Computer Vision

Module number	CS 717
Title	Master Seminar on Computer Vision
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Higher Level Computer Vision, Generative Computer Vision Models or Image Processing
Aim of module	The student prepares a scientific paper and gives a presentation on a current research topic based on published research. State-of-the-art topics are proposed by the professors. The paper and the presentation are prepared under the guidance of a professor or a research staff member. Active participation in the seminar discussions is expected
Learning outcomes and qualification goals	Professional competence: • The student gains a deep understanding of the research topic • He/she is able to describe/summarize the topic in detail in his/her own words • He/she reflects on the topic and judges the contribution of the research papers Methodological competence: • The student is able to write a well-structured scientific paper and to present his/her results • He/she is also aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar Personal competence: • The student has learned how to write a well-structured, concise paper and give a presentation • This is part of the preparation to write and present a master's thesis
Media	Scientific papers and books
Literature	Depending on the topic of the seminar
Methods	Do scientific reading independently under the guidance of a professor or a research staff member, active discussions in a group of peers
Form of assessment	• 50% Presentations (one 3-5 minutes presentation and one 25-30 minutes presentation) • 50% Seminar Report (10-20 pages, has to be submitted after three month)
Admission requirements for assessment	-

Duration of assessment	N/A
Language	English
Offering	HWS/FSS
Person in charge	Prof. Dr.-Ing. Margret Keuper
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	3rd semester

CS 720: Uncertainty Estimation

Module number	CS 720
Title	Uncertainty Estimation
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Bachelor's degree, the fundamentals
Aim of module	In this seminar, students perform scientific research, either in the form of a literature review or by conducting a small experiment, or a mixture of both, and prepare a written report about the results. Topics of interest focus around a variety of problems and tasks from the fields of Data Mining, Web Mining, or the Semantic Web
Learning outcomes and qualification goals	Professional competence: • Students will acquire a deep understanding of the research topic • He/she is expected to describe in-depth and summarize the topic in detail in his/her own words, as well as to judge the contribution of the research papers to ongoing research (MK1, MK2, MK3) Methodological competence: • Students will develop methods and skills to find relevant literature for his/her topic, to prepare methodologically sound scientific experiments, and to write a well-structured scientific paper and to present his/her results • He/she will also be aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar (MF2) Personal competence: • Students will acquire skills on how to find relevant literature for a research topic, organize a small research task, write a wellstructured, concise paper about it and present the results of their work • He/she is well prepared to write and present a master's thesis (MF2)
Media	Scientific papers and books
Literature	Depends on the topic of the seminar
Methods	Do scientific work independently under the guidance of a professor or a research staff member
Form of assessment	• Seminar paper (5-25 pages depending on the topic) • Peer Review (1-10 pages depending on the topic) • Presentation (15-60 minutes depending on the topic)
Admission requirements for assessment	-

Duration of assessment	• 6-8 weeks (writing of the report) • 15-60 minutes (presentation)
Language	English or German
Offering	Irregular
Person in charge	Prof. Dr. Heiko Paulheim
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	3rd semester

CS 721: Seminar Data-Science I

Module number	CS 721
Title	Seminar Data-Science I
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	There are no formal requirements. However, previous participation in the courses “Network Science” and “Text Analytics” are recommended
Aim of module	In this seminar, students perform scientific research, either in the form of a literature review or by conducting a small experiment, or a mixture of both, and prepare a written report about the results. Topics of interest focus around a variety of problems and tasks from the fields of Data-Science, Network Science and Text Mining
Learning outcomes and qualification goals	Professional competence: • Students will acquire a deep understanding of the research topic • He/she is expected to describe in-depth and summarize the topic in detail in his/her own words, as well as to judge the contribution of the research papers to ongoing research (MK1, MK2, MK3) Methodological competence: • Students will develop methods and skills to find relevant literature for his/her topic, to prepare methodologically sound scientific experiments, and to write a well-structured scientific paper and to present his/her results • He/she will also be aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar (MF2) Personal competence: • Students will acquire skills on how to find relevant literature for a research topic, organize a small research task, write a wellstructured, concise paper about it and present the results of their work • He/she is well prepared to write and present a master’s thesis (MF2)
Media	Slides, scientific papers, blackboard (electronic)
Literature	Depends on the topic
Methods	Self-study of assigned material, presentation of scientific articles, joint discussion of work, collaboration with peers
Form of assessment	• Preparation of a seminar report (12-14 pages, plus bibliography) • Presentation of the seminar paper and answering questions (40 minutes) • Participation in the presentation discussions • Reflection on a seminar report written by someone else

Admission requirements for assessment	-
Duration of assessment	-
Language	English
Offering	Irregular
Person in charge	Prof. Markus Strohmaier
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	3rd semester

CS 733: Advanced Topics in Process Mining

Module number	CS 733
Title	Advanced Topics in Process Mining
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Participants are expected to have fundamental knowledge of process mining, including basic concepts, techniques, and tools, to actively engage with the seminar content
Aim of module	In this seminar, each student conducts scientific research on advanced topics in process mining, either as a literature review, a small experimental study, or a combination of both. Each student prepares a written report and delivers a presentation to the class, sharing their findings and insights with peers. Topics of interest include classical process mining tasks, such as process discovery, conformance checking, and predictive process monitoring, explored through current research trends and novel approaches, including generative AI, machine learning applications, and the analysis of non-traditional event data types. The seminar encourages students to investigate innovative methods, emerging challenges, and practical implications in the field
Learning outcomes and qualification goals	Professional competence: • Students will develop a deep understanding of the research topic • They are expected to critically analyze and comprehensively summarize the topic in their own words, as well as evaluate the contribution of the research papers to ongoing research in the field Methodological competence: • Students will develop the methods and skills needed to identify relevant literature for their topic, design methodologically sound scientific experiments, write a well-structured scientific paper, and effectively present their results • They will also gain awareness of the importance of academic integrity and the need to avoid plagiarism Personal qualification: • Students will acquire the skills required to identify relevant literature for a research topic, organize and conduct a small-scale research project, write a well-structured and concise academic paper, and effectively present their results • They will be well prepared to write and present a Master's thesis
Media	Scientific papers and books

Literature	• Depends on the specific topic • For a general overview of the field, the following book is recommended: Wil van der Aalst: <i>Process Mining: Data Science in Action</i>, 2nd edition
Methods	Perform scientific work independently under the guidance of a professor or a research staff member
Form of assessment	• Seminar paper (50%) • Mid-term presentation (10%) • Final presentation (40%)
Admission requirements for assessment	-
Duration of assessment	• Seminar paper: 18 pages (excluding appendix and references) • Mid-term presentation: 10 minutes presentation + 10 minutes discussion • Final presentation: 20 minutes presentation + 10 minutes discussion
Language	English
Offering	Irregular
Person in charge	Prof. Dr. Daniel Schuster
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	3rd semester

CS 735: Computational Analysis of Communication

Module number	CS 735
Title	Computational Analysis of Communication
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	Hours in presence: 28 h per semester (2 SWS) Self-study: 92 h per semester, including • 45 h: Lectures/exercises • 47 h: Research report
Prerequisites	Basic skills in descriptive and inferential statistics, basic knowledge of data structures and data wrangling procedures, machine learning, web-scraping/web-mining
Aim of module	Computational methods give us new tools to investigate the content of communication as well as its effect at a larger scale than previously possible. Old questions – which topics are currently salient? How are they framed? – can be investigated for larger time frames and with larger samples, while technological change opens up new questions – for example, how new sources of information, e.g. disinformation spreaders on one hand and citizen journalists on the other, affect public communication. Questions like these revolve around the process of communication. Sound research in this area thus requires both a solid foundation from communication theory as well as expertise in handling new and “big” data. To close this gap, the growing discipline of Computational Communication Science (CCS) takes on a combinatory perspective between social and computer science. The present course will provide an overview of the current state of CCS and intends to motivate students to approach pressing social questions from a different perspective. A specific focus will be on the application of communications methods skills – assessing and ensuring validity and reliability of measures through establishing routines for refining your operationalization and improving your measurement to capture theoretically rich concepts – to semisupervised and automated methods
Learning outcomes and qualification goals	Professional competence: • After the course the students are aware of the typical research topics and questions in automated media content analyses and the different methodological approaches for tackling them • They are able to operationalize specific concepts and understand how to assess the validity and reliability of their measurement tools • They know the different methods’ potentials, limitations, and typical fields of application • They are able to develop their own specific research questions and can make an informed decision about which method to apply for answering it (MK3)

	Methodological competence: • Students are able to independently develop a research question and design in the area of automated media content analysis and can conduct a respective analysis using one of the different methodological approaches introduced in the exercise • They are able to document the results of their analyses in a research report and reflect upon their findings' limitations with regards to reliability and validity (MF1, MF2, MF3, MF4, MF5) Personal competence: • The course supports students to develop problem-solving competences with regards to research-design oriented questions • By solving exercises independently, the transfer of the learned material to related questions is promoted and self-confidence with regards to research-oriented tasks is gathered (MKO3)
Media	Exercise sheets and lecture slides are available online
Literature	van Atteveldt, W., Trilling, D., & Arcila, C. (2021). <i>Computational Analysis of Communication: A practical introduction to the analysis of texts, networks, and images with code examples in Python and R</i> . http://cssbook.net/
Methods	Lecture elements, student presentations, weekly exercises, literature studies
Form of assessment	• Written research report (10-12 pages) • Presentation (15 minutes)
Admission requirements for assessment	-
Duration of assessment	4-8 weeks, depending on the topic
Language	English
Offering	HWS/FSS
Person in charge	Dr. Rainer Freudenthaler
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science, M.Sc. Social Data Science
Semester	1st/ 2nd/ 3rd semester

IE 704: Seminar AI Systems Engineering

Module number	IE 704
Title	Seminar AI Systems Engineering
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	-
Aim of module	The student prepares a scientific paper and gives a presentation on a current research topic based on published research. State-of-the-art topics are proposed by the supervisors. The paper and the presentation are prepared under the guidance of a research staff member. Active participation in the seminar discussions is expected
Learning outcomes and qualification goals	Professional competence: • The student gains a deep understanding of the research topic • He/she is able to describe/summarize the topic in detail in his/her own words • He/she reflects on the topic and judges the contribution of the research papers (MK1, MK2, MK3) Methodological competence: • The student is able to write a well-structured scientific paper and to present his/her results • He/she is also aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar (MF2) Personal competence: • The student has learned how to write a well-structured, concise paper and give a presentation • This is part of the preparation to write and present a master's thesis (MF2)
Media	Scientific papers and books, final presentation with PowerPoint
Literature	Depends on the topic of the seminar
Methods	Do scientific work independently under the guidance of a professor or a research staff member, active discussions in a group of peers
Form of assessment	• 25% Reviews and Discussion (Review: 1-3 pages) • 25% Presentation (15 minutes presentation and 10 minutes discussion) • 25% Seminar paper submitted for review (max 20 pages in Springer proceedings format) • 25% "Camera-ready" seminar paper (max 20 pages in Springer proceedings format)

Admission requirements for assessment	-
Duration of assessment	Writing time for papers and reviews 4 to 8 weeks
Language	English
Offering	Irregular
Person in charge	Dr. Christian Bartelt
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	3rd semester

H. Master Thesis

Master Thesis

Title	Master's Thesis
Form of module	Master's Thesis
Type of module	Thesis
Level	Master
ECTS	30
Workload	Self study: 840 h per semester
Prerequisites	-
Aim of Modules	Develop a deep understanding of an advanced topic of data science
Learning outcomes and qualifications goals	Professional competence: • The student has a deep understanding of an advanced topic (MK1) Methodological competence: • The student is familiar with methods for analysing and independently solving advanced, complex problems (MK1, MK2, MK3) Personal competence: • The student has the capability to understand, analyse and independently find solutions to advanced, complex problems • The student has the capability to assess and understand the state-of-the-art in business informatics and adapt the latest technologies and methods to solve real world problems • The student is able to present a complex topic in written and oral form in a clear and understandable way (MF1, MF2, MF3, MF4, MKO2, MKO3)
Media	Various
Literature	Topic dependent
Methods	Independent research work
Form of assessment	Written thesis, 10-120 pages in length depending on the topic
Admission requirements for assessment	To be permitted to write the master thesis, the student is to obtain at least 60 ECTS; For students subject to the 2024 examination regulations, it is additionally required that the Data Science Fundamentals area has been successfully completed
Duration of assessment	6 months
Languages	English only
Offering	HWS/FSS

Person in charge	Examiners: University teachers, auxiliary professors, honorary professors and senior academic staff members of the School of Business Informatics and Mathematics or of the School of Social Sciences. Prof. Strohmaier. Supervisors of other faculties are possible upon application to the examination board and after examination of the content
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science
Semester	4th semester

Abbreviations

Knowledge

This degree program provides students with a solid theoretical foundation as well as practical skills for data management, data analytics methods, responsible data science and data science applications. The courses are divided in five groups - Data Science Fundamentals, Data Management, Data Analytics Methods, Responsible Data Science, and Data Science Applications. After studying the fundamentals of Data Science, students will then learn how to manage, analyse, and responsibly handle data. Furthermore, they will learn about the different applications of data science in computer science, social sciences, or business and economics. In addition to the regular lecture courses, students participate in a one or two semester team project or individual project.

During their studies

- (MK1) all students develop a deep understanding of the relevant concepts, methods and problem-solving strategies used in different application domains;
- (MK2) technology-oriented students learn the concepts, algorithms and strategies used to solve concrete, practical application-oriented problems in informatics;
- (MK3) social sciences-oriented students develop a deep understanding of how to set up, analyse and interpret advanced empirical research questions.

As part of this education, students become familiar with a wide range of models, modelling languages, methods, and tools. Regardless of their specialization, students also learn how to collect, structure, manipulate, prepare, interpret, communicate, and use data, information and knowledge.

Capabilities

After completing their studies, students have the ability to

- (MF1) apply a wide range of abstraction and analysis techniques;
- (MF2) understand, interpret, describe and present relevant scientific publications;
- (MF3) exploit the latest scientific results;
- (MF4) independently tackle problems in data management and analytics and describe their results in a structured, written form;
- (MF5) continue their studies at the PhD level, if their results are of sufficient quality.

Competencies

After completing their studies, students have the competences needed to

- (MKO1) apply their knowledge and capabilities to solve specific problems in a team context;
- (MKO2) use their interdisciplinary education to mediate between technical and non-technical individuals;
- (MKO3) evaluate the latest changes in programming languages, systems, models and, wherever possible, exploit them to develop better solutions to data-science related problems.